Quantum Braid Dynamics

A Computational Process

R. Fisher

October 30, 2026

Contents

Part 6: Appendices **3**

1. **Abramsky, S., Barbosa, R. S., & Searle, A. (2024).** 3
2. **Acharya, R., et al. (2024).** 3
3. **Adams, R. A., & Fournier, J. J. (2003).** 3
4. **Aleksandrowicz, G., et al. (2019).** 4
5. **Ambjorn, J., Jurkiewicz, J., & Loll, R. (2005).** 4
6. **Anderson, E. (2012).** 4
7. **Ashtekar, A. (1986).** 4
8. **Awodey, S. (2010).** 5
9. **Baader, F., & Nipkow, T. (1998).** 5
10. **Barbour, A. D., Holst, L., & Janson, S. (1992).** 5
11. **Bekenstein, J. D. (1981).** 6
12. **Belkin, M., & Niyogi, P. (2003).** 6
13. **Bennett, C. H. (1982).** 6
14. **Bollobas, B. (2001).** 6
15. **Bombelli, L., Lee, J., Meyer, D., & Sorkin, R. D. (1987).** 7
16. **Bondy, J. A., & Murty, U. S. R. (2008).** 7
17. **Calder, J., & Garcia Trillos, N. (2022).** 7
18. **Carlip, S. (2009).** 8
19. **Cheeger, J., Colding, T. H., & Tian, G. (1997).** 8
20. **Coleman, S. (1977).** 8
21. **Dauphinais, G., Kribs, D. W., & Vasmer, M. (2024).** 8
22. **Diestel, R. (2017).** 9
23. **Dowker, F. (2005).** 9
24. **Ehrig, H., Ehrig, K., Prange, U., & Taentzer, G. (2006).** 9
25. **Enderton, H. B. (2001).** 10
26. **Erdos, P., & Renyi, A. (1960).** 10
27. **Fuchs, C. A. (2010).** 10
28. **Gambini, R., Porto, R. A., Torterolo, S., & Pullin, J. (2009).** 10
29. **Gilbarg, D., & Trudinger, N. S. (2001).** 11
30. **Gillespie, D. T. (1977).** 11
31. **Godel, K. (1931).** 11
32. **Gorard, J. (2020).** 12
33. **Gottesman, D. (1997).** 12
34. **Gukov, S., Takayanagi, T., & Toumbas, N. (2004).** 12
35. **Harlow, D. (2016).** 12
36. **Hawking, S. W., & Ellis, G. F. R. (1973).** 13
37. **Hossenfelder, S. (2011).** 13

38. **Jacobson, T. (1995).** . . . 13
39. **Jacobson, T. (2016).** . . . 14
40. **Janson, S. (1987).** . . . 14
41. **Jaynes, E. T. (1957).** . . . 14
42. **Jones, V. F. R. (1985).** . . . 15
43. **Jost, J., & Liu, S. (2016).** . . . 15
44. **Kitaev, A. Y. (2003).** . . . 15
45. **Kobakhidze, A. (2011).** . . . 15
46. **Lamport, L. (1978).** . . . 16
47. **Landauer, R. (1991).** . . . 16
48. **Leibniz-Clarke Correspondence (1715-1716).** . . . 16
49. **Liggett, T. M. (1999).** . . . 17
50. **Maldacena, J. M. (1998).** . . . 17
51. **Marker, D. (2002).** . . . 17
52. **Mousa, M., Jamadagni, A., et al. (2025).** . . . 17
53. **Ollivier, Y. (2009).** . . . 18
54. **Otto, F., Mansuroglu, R., Schuch, N., Guhne, O., & Sahlmann, H. (2025).** . . . 18
55. **Padmanabhan, T. (2009).** . . . 18
56. **Page, D. N. (1993).** . . . 19
57. **Page, D. N., & Wootters, W. K. (1983).** . . . 19
58. **Palmigiano, A., & Sadrzadeh, M. (Eds.). (2023).** . . . 19
59. **Pastawski, F., Yoshida, B., Harlow, D., & Preskill, J. (2015).** . . . 19
60. **Pemantle, R. (1992).** . . . 20
61. **Penington, G. (2019).** . . . 20
62. **Rodrigues, F. L. S., & Lutz, E. (2025).** . . . 20
63. **Rovelli, C. (1996).** . . . 21
64. **Rovelli, C., & Smolin, L. (1990).** . . . 21
65. **Ryu, S., & Takayanagi, T. (2006).** . . . 21
66. **Sachs, H. (1962).** . . . 21
67. **Sakharov, A. D. (1967).** . . . 22
68. **Sati, H., & Schreiber, U. (2025).** . . . 22
69. **Singer, A., & Wu, H.-T. (2013).** . . . 22
70. **Sorkin, R. D. (2005).** . . . 23
71. **Steinberg, M. et al. (2025).** . . . 23
72. **Uustalu, T., & Vene, V. (2008).** . . . 23
73. **Vaccaro, J. A., & Barnett, S. M. (2011).** . . . 23
74. **van der Hoorn, P., & Stegehuis, C. (2020).** . . . 24
75. **van Kampen, N. G. (1992).** . . . 24
76. **van Luxburg, U., Belkin, M., & Bousquet, O. (2008).** . . . 24
77. **Verlinde, E. (2011).** . . . 25
78. **Wang, W. (2024).** . . . 25
79. **Wheeler, J. A. (1990).** . . . 25
80. **Wilson, K. G. (1975).** . . . 25
81. **Witten, E. (1989).** . . . 26
82. **Woess, W. (2000).** . . . 26
83. **Wolfram, S. (2002).** . . . 26
84. **Wolfram, S. (2020).** . . . 26
85. **Zurek, W. H. (2003).** . . . 27

GitHub Repository: https://github.com/braiddynamics/qbd-portal . . . 27

Nomenclature & Symbol Table **27**

Document Status . . . 47

Part 6: Appendices

1. Abramsky, S., Barbosa, R. S., & Searle, A. (2024).

"Combining contextuality and causality: a game semantics approach" - *Philosophical Transactions of the Royal Society A*, 382(2268), 20230002 * **Link:** https://royalsocietypublishing.org/doi/10.1098/rsta.2023.0002

Overview: Abramsky, Barbosa, and Searle develop a unified structure that combines quantum contextuality and discrete causality using game semantics. They show how these semantic structures capture the non-local correlation limits of quantum systems while preserving the strict causal ordering of events, establishing a precise logic for relativistic quantum networks.

Relevance to QBD: This semantic structure is vital for the causal quantum models formulated in Chapter 14. To show that the non-local correlations of our topological qubits do not violate discrete causality, we analyze them using Abramsky's game semantics. Abramsky's results supply the bounds needed to confirm that QBD is both contextual and causally consistent.

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2. Acharya, R., et al. (2024).

"Bridging classical and quantum: Group-theoretic approach to quantum circuit simulation" - *Physical Review Letters*, 132(15), 150602 * **Link:** https://journals.aps.org/prl/abstract/10.1103/PhysRevLett.132.150602

Overview: Acharya and collaborators develop a unified algebraic formalism that maps quantum circuit states and gates to representation-theoretic structures in finite group theory. By leveraging group-theoretic structures, they show that specific classes of quantum circuits, such as stabilizer circuits and their near-stabilizer extensions, can be mapped onto classical group actions. This work pins down exact boundary conditions for when quantum resources yield computational advantages over classical simulations.

Relevance to QBD: Within Quantum Braid Dynamics, this algebraic mapping is pivotal for formalizing how topological excitations acting as qubits under the stabilizer formalism correspond to discrete representations of the braid group. This reference validates that the discrete stabilizers and gates formed by braid permutations translate directly to classical and quantum group-theoretic actions on the causal graph. Drawing on this result, we can show why the discrete update rule naturally constructs a universal quantum computer without the need for continuous fields.

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3. Adams, R. A., & Fournier, J. J. (2003).

"Sobolev Spaces" * **Link:** https://www.sciencedirect.com/book/9780120441433/sobolev-spaces

Overview: Adams and Fournier present a comprehensive and classic monograph on the theory of Sobolev spaces. They cover the fundamental properties of these spaces, including approximation theorems, embedding theorems, and compactness results. Their work supplies the analytical machinery used to analyze partial differential equations on continuous domains.

Relevance to QBD: This reference is indispensable for the continuum limit derivations of QBD. In Chapter 12, the discrete graph Laplacian and its associated energy functionals are proved to converge to continuous differential operators. This convergence requires mapping graph functions to Sobolev spaces. The embedding theorems derived by Adams and Fournier provide the required bounds to ensure that the discrete solutions remain well-behaved as the graph spacing approaches zero.

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4. Aleksandrowicz, G., et al. (2019).

"Qiskit: An Open-source Framework for Quantum Computing" * **Link:** https://zenodo.org/record/2562111

Overview: Aleksandrowicz and the Qiskit team describe the architecture and implementation of Qiskit, an open-source software suite designed for writing, compiling, and running quantum circuits. The platform provides the software layers required to translate high-level quantum algorithms into the precise physical controls needed to execute circuits on real quantum hardware or classical simulators.

Relevance to QBD: QBD leverages the stabilizer code model to protect topological graph structures from vacuum noise. In Chapter 10, Qiskit is utilized to simulate and verify the fault-tolerant operations of the tripartite braid qubits. This reference anchors the software standard used to model the stabilizer generators and logical gates, bridging the gap between high-level topological code theory and numerical circuit validation.

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5. Ambjorn, J., Jurkiewicz, J., & Loll, R. (2005).

"Reconstructing the Universe" * **Link:** https://arxiv.org/abs/hep-th/0505154

Overview: Ambjorn, Jurkiewicz, and Loll demonstrate that a non-trivial four-dimensional classical spacetime can emerge from a non-perturbative path integral of causal triangulations. This approach, known as Causal Dynamical Triangulations (CDT), shows that imposing a strict distinction between space-like and time-like steps solves the historical problem of spatial collapse and ensures causality in the continuum limit.

Relevance to QBD: This seminal work in discrete quantum gravity provides vital conceptual backing for the geometrogenesis proofs in QBD. In Chapter 11, we leverage Loll's insights to show how discrete causal structures avoid cosmological dimensional collapse. CDT's results set a precedent for how discrete, causally ordered structures can successfully yield continuous, high-dimensional geometries when the continuum limit is taken.

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6. Anderson, E. (2012).

"The Problem of Time in Quantum Gravity" * **Link:** https://arxiv.org/abs/1009.2157

Overview: Anderson provides a thorough review of the problem of time in canonical quantum gravity, a conceptual crisis arising because general relativity is a reparametrization-invariant theory with no preferred background clock. He explores various proposed solutions, including relational time, the Page-Wootters mechanism, and semiclassical approximations.

Relevance to QBD: The problem of time is resolved in QBD by the dual-time architecture developed in Chapter 14. By separating logical time, which counts rewrite steps, from emergent physical time, which corresponds to the length of causal paths, we bypass the need for a background clock. Anderson's review situates this resolution within the broader literature and contrasts our discrete causal ordering against the difficulties encountered by continuous canonical formulations.

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7. Ashtekar, A. (1986).

"New Variables for Classical and Quantum Gravity" * **Link:** https://journals.aps.org/prl/abstract/10.1103/PhysRevLett.57.2244

Overview: Ashtekar introduces a new set of canonical variables for general relativity, mapping the theory's phase space to that of a Yang-Mills gauge theory. This reformulation simplifies the constraints of classical

gravity and lays the foundation for Loop Quantum Gravity by expressing the spatial geometry in terms of loops and connections.

Relevance to QBD: Ashtekar variables provide the direct inspiration for how spatial geometry can be encoded in connection-like loops within a discrete graph. In Chapter 13, the discrete field equations are formulated by defining connection variables along the edges of the causal graph. Ashtekar's treatment of canonical gravity traces the historical and algebraic link between continuous general relativity and our discrete, loop-based gauge formulations.

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8. Awodey, S. (2010).

"Category Theory (2nd ed.)" * **Link:** https://global.oup.com/academic/product/category-theory-9780199237180

Overview: Awodey provides a systematic and accessible introduction to category theory, focusing on the structures and relations that unify different branches of mathematics. He covers functors, natural transformations, limits, colimits, and monads, emphasizing the structural perspective over set-theoretic foundations.

Relevance to QBD: Category theory is the formal language used to define the computational syntax of QBD in Chapter 1. By representing the substrate as a category where vertices are objects and edges are morphisms, we formalize the rewrite rules as functors. Awodey's text is the core reference for these category-theoretic structures, ensuring that the algebraic foundations of our model are carefully defined.

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9. Baader, F., & Nipkow, T. (1998).

"Term Rewriting and All That" * **Link:** http://dx.doi.org/10.1017/CBO9781139172752

Overview: Baader and Nipkow present a comprehensive guide to the theory of term rewriting systems. They cover abstract reduction systems, confluence, termination, and unification. Their work documents the core logical principles that govern how symbolic expressions can be systematically modified under a set of deterministic rewrite rules.

Relevance to QBD: QBD operates as a discrete dynamical system driven by graph rewriting. In Chapter 2, we prove that the update rule is confluent and terminating within the causal horizon, ensuring that physical history is unique and well-defined. Appealing to Baader and Nipkow supplies the logical tools required for this confluence proof, showing that our local rewrite rules behave as a consistent term rewriting system.

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10. Barbour, A. D., Holst, L., & Janson, S. (1992).

"Poisson Approximation" - *Oxford University Press* * **Link:** https://global.oup.com/academic/product/poisson-approximation-9780198522355

Overview: Barbour, Holst, and Janson present a detailed study of the Poisson approximation, utilizing Stein's method to derive explicit error bounds for the approximation of independent or weakly dependent random variables. Their work delivers exact tools for analyzing the statistical properties of rare events in complex systems.

Relevance to QBD: In Chapter 5, we analyze the statistical distribution of minimal cycles in the vacuum graph. Because these cycles are rare, their distribution is modeled using the Poisson approximation. The precise error bounds developed by Barbour and his co-authors are used to prove that the vacuum converges to a sparse, stable state, grounding the claim that the background spacetime remains flat and stable.

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11. Bekenstein, J. D. (1981).

"A universal upper bound on the entropy-to-energy ratio for bounded systems" * **Link:** https://journals.aps.org/prd/abstract/10.1103/PhysRevD.23.287

Overview: Bekenstein derives the universal upper bound on the entropy-to-energy ratio for any thermodynamic system of localized energy confined within a sphere of effective radius. By analyzing the thermodynamic limits of black hole absorption, this work proves that the information capacity of a physical system is strictly bounded by its spatial boundary and energy content, confirming information as a foundational constraint on relativistic thermodynamics.

Relevance to QBD: The Bekenstein Bound serves as a central physical selection rule in QBD, represented by the Bekenstein Bound Lemma in Chapter 16. It motivates why the vacuum rewrite rule suppresses high-density graph fluctuations that exceed the information density limit through the quadratic deletion flux. The bound is structurally enforced because a region of graph space cannot support more independent cycles than its vertex horizon allows, formalizing the emergence of holographic screen mechanisms.

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12. Belkin, M., & Niyogi, P. (2003).

"Laplacian Eigenmaps for Dimensionality Reduction and Data Representation" * **Link:** https://www2.imm.dtu.dk/projects/manifold/Papers/Laplacian.pdf

Overview: Belkin and Niyogi introduce Laplacian Eigenmaps, a geometric toolset that utilizes the Laplace-Beltrami operator to map high-dimensional data points to a low-dimensional manifold while preserving local proximity. They prove that the eigenvectors of the graph Laplacian provide optimal embeddings that preserve the underlying manifold's geometry.

Relevance to QBD: This approach is the foundation for the dimensional reconstruction proofs in Chapter 12. To show that a discrete graph converges to a smooth spacetime manifold, we must construct an embedding. Belkin and Niyogi's work provides the justification for using the graph Laplacian's eigenvectors to recover the metric tensor and coordinate charts, demonstrating that low-dimensional spacetime emerges naturally from discrete networks.

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13. Bennett, C. H. (1982).

"The thermodynamics of computation: a review" * **Link:** https://link.springer.com/article/10.1007/BF02084158

Overview: Bennett reviews the thermodynamics of computation, focusing on the relation between logical reversibility and physical dissipation. He clarifies Landauer's principle, proving that while logical operations themselves do not necessarily require energy dissipation, the erasure of information or the resetting of memory registers is always accompanied by a physical entropy increase.

Relevance to QBD: Bennett's insights are foundational for the dynamical rewrite rules formulated in Chapter 4. The update engine behaves as a computational constructor that deletes and instantiates edges. The physical cost of these updates is governed by Bennett's thermodynamic limits, ensuring that information erasure at the graph level generates localized heat. This couples the computational activity of the universe directly to thermodynamic energy.

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14. Bollobas, B. (2001).

"Random Graphs (2nd ed.)" * **Link:** https://doi.org/10.1017/CBO9780511814068

Overview: Bollobas presents a classic and detailed monograph on the theory of random graphs, focusing on the probabilistic methods used to study the properties of graphs generated by random processes. He covers connectivity, path lengths, chromatic numbers, and the threshold functions that govern the appearance of specific subgraphs.

Relevance to QBD: This reference is integral to the random graph audits conducted in Chapter 5. To prove that the vacuum graph remains sparse and does not undergo runaway densification, we must analyze the threshold behavior of its local connections. Bollobas's probabilistic bounds provide the disciplined apparatus required to analyze the stability of the vacuum against runaway graph growth.

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15. Bombelli, L., Lee, J., Meyer, D., & Sorkin, R. D. (1987).

"Space-time as a causal set" * **Link:** https://journals.aps.org/prl/abstract/10.1103/PhysRevLett.59.521

Overview: Bombelli and his collaborators introduce the causal set approach to quantum gravity, postulating that spacetime is a discrete partially ordered set (poset) where the partial order represents causal relations. They show that discrete causal structure is sufficient to recover both the causal structure and the local volume of a smooth spacetime manifold.

Relevance to QBD: This classic paper is the conceptual precursor to the Causal Graph substrate defined in Chapter 1. We adopt Sorkin's insight that causality is fundamental and volume is discrete. However, we expand this setting by adding relational graph connectivity, which allows the modeling of quantum state vector updates. This citation places QBD within the historical and physical lineage of discrete causality as the chief organizer of spacetime.

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16. Bondy, J. A., & Murty, U. S. R. (2008).

"Graph Theory" * **Link:** https://link.springer.com/book/9781846289699

Overview: Bondy and Murty present a modern, graduate-level textbook on graph theory. They cover connectivity, matchings, independent sets, graph colorings, and topological graph theory. Their work provides a thorough and comprehensive collection of tools used to analyze discrete relational networks.

Relevance to QBD: This textbook serves as the standard reference for all graph-theoretic operations conducted across the monograph. Whether analyzing the path length between vertices in Chapter 11 or evaluating the cycle structure of braids in Chapter 6, the theorems and notations of Bondy and Murty ensure that our discrete derivations are mathematically sound.

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17. Calder, J., & Garcia Trillos, N. (2022).

"Improved spectral convergence rates for graph Laplacians on epsilon-graphs and k-NN graphs" * **Link:** https://arxiv.org/abs/1910.13476

Overview: Calder and Garcia Trillos derive improved spectral convergence rates for graph Laplacians converging to continuous Laplace-Beltrami operators on data-generated manifolds. They utilize optimal transport theory and variational analysis to prove that the eigenvectors and eigenvalues of the discrete graph match their continuous counterparts with sharp convergence bounds.

Relevance to QBD: This variational analysis is decisive for the continuum limit of the discrete field equations in Chapter 12. To show that the discrete Einstein-Hilbert action on our graph converges to the continuous action, we must bound the error of the graph Laplacian. Calder's convergence bounds are used to confirm that the discrete curvature converges systematically to the continuous Ricci scalar, establishing a bridge to classical general relativity.

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18. Carlip, S. (2009).

"Spontaneous dimensional reduction in short-distance quantum gravity?" - *AIP Conference Proceedings*, 1196(1), 72-78. * **Link:** https://doi.org/10.1063/1.3284402

Overview: Carlip reviews evidence across diverse quantum gravity approaches (causal dynamical triangulations, loop quantum gravity, asymptotically safe gravity, and Ho?ava-Lifshitz gravity) indicating that spacetime dimension spontaneously reduces to $d \approx 2$ near the Planck scale.

Relevance to QBD: Carlip's phenomenological synthesis provides key context for the spectral dimension flow observed in Chapter 5, confirming that the short-scale reduction to tree-like ($d \sim 2$) pre-geometry before infrared dimensional expansion is a generic, physically expected feature of discrete quantum gravity models.

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19. Cheeger, J., Colding, T. H., & Tian, G. (1997).

"On the singularities of spaces with bounded Ricci curvature" * **Link:** https://www.semanticscholar.org/paper/On-the-singularities-of-spaces-with-bounded-Ricci-Cheeger-Colding/9b384c019d715a63e6a34b2296412c3e4c4ded84

Overview: Cheeger, Colding, and Tian analyze the structure of singularities in limit spaces of Riemannian manifolds with bounded Ricci curvature. They prove that these limit spaces, though singular, possess tightly constrained geometric properties, specifically regarding their tangent cones and the Hausdorff dimension of their singular sets.

Relevance to QBD: In Chapter 13, we must analyze the singular behavior of the discrete geometry when local graph densities fluctuate. Cheeger's analysis of singular limit spaces is used to prove that the emergent discrete spacetime remains stable and does not develop uncontrollable geometric singularities, ensuring that physical observables remain finite and well-defined even at the smallest scales.

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20. Coleman, S. (1977).

"The Uses of Instantons" * **Link:** http://www.physics.mcgill.ca/~jcline/742/Coleman-Instantons.pdf

Overview: Coleman presents a set of lectures on the role of instantons, which are classical solutions to the equations of motion in Euclidean spacetime. He explains how these non-perturbative configurations correspond to quantum tunneling events between different vacuum states, documenting the physical basis for non-abelian gauge vacuum structure.

Relevance to QBD: Instantons are the continuous analogs of the non-perturbative transition operations that drive gauge dynamics in Chapter 8. In QBD, the tunneling of a tripartite braid between different topological phases corresponds to a discrete instanton-like event in the causal history. Coleman's lectures are cited to draw this physical analogy, grounding why non-abelian gauge structures emerge from topological updates.

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21. Dauphinais, G., Kribs, D. W., & Vasmer, M. (2024).

"Stabilizer Formalism for Operator Algebra Quantum Error Correction" * **Link:** https://quantum-journal.org/papers/q-2024-02-21-1261/pdf

Overview: Dauphinais, Kribs, and Vasmer generalize the stabilizer formalism from finite-dimensional quantum systems to infinite-dimensional operator algebras. They develop a careful algebraic treatment that per-

mits stabilizer codes to be defined on operator algebras, proving that the standard error correction conditions are preserved under this algebraic generalization.

Relevance to QBD: This algebraic apparatus is indispensable for formalizing topological protection in QBD. In Chapter 10, the tripartite braid quantum states are shown to reside within a stable, topologically protected codespace. This work provides the backing required to define stabilizer operators on the infinite-dimensional algebra of our emergent graph, ensuring that the quantum information remains protected from local graph perturbations.

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22. Diestel, R. (2017).

"Graph Theory (5th ed.)" - *Springer* * **Link:** https://diestel-graph-theory.com/

Overview: Diestel presents a detailed and standard textbook on graph theory. The author covers infinite graphs, graph limits, topological aspects of graphs, and the structural properties that emerge in large-scale networks. The book serves as the leading reference for advanced graph-theoretic structures.

Relevance to QBD: This textbook is the foundation for the graph-theoretic proofs across the monograph. In Chapter 11, we utilize Diestel's theorems on infinite graphs to formulate the infinite-volume limit of our causal network. The connection to these results confirms that the discrete graph structures remain mathematically consistent and well-defined even when the number of vertices approaches infinity, paving the way for the continuous spacetime manifold.

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23. Dowker, F. (2005).

"Causal sets and the deep structure of spacetime" * **Link:** https://arxiv.org/abs/gr-qc/0508109

Overview: Dowker provides a conceptual and physical overview of the causal set approach to quantum gravity. She argues that spacetime is fundamentally discrete and that the continuum is merely an approximation. The author demonstrates that discrete causal sets successfully preserve Lorentz invariance, solving a major historical challenge faced by discrete models.

Relevance to QBD: Dowker's work is a key conceptual pillar for the discrete causal substrate defined in Chapter 1. We adopt her insight that discrete causal ordering is sufficient to construct macroscopic geometry. In Chapter 14, we prove that QBD preserves Lorentz covariance in the continuum limit, invoking Dowker to show why our discrete causal steps naturally satisfy relativistic constraints.

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24. Ehrig, H., Ehrig, K., Prange, U., & Taentzer, G. (2006).

"Fundamentals of Algebraic Graph Transformation" - *Monographs in Theoretical Computer Science. An EATCS Series, Springer* * **Link:** https://doi.org/10.1007/3-540-31188-2

Overview: Ehrig et al. formulate the foundational theory of algebraic graph transformation using category theory (adhesive and quasi-adhesive categories, double-pushout approach). They prove critical pair confluence and local Church-Rosser theorems for graph rewriting systems.

Relevance to QBD: The double-pushout framework and graph confluence theorems are used in Chapter 2 to establish the confluence, causality, and causal consistency of local rewrite rules operating on the relational substrate.

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25. Enderton, H. B. (2001).

"A Mathematical Introduction to Logic (2nd ed.)" * **Link:** https://www.sciencedirect.com/book/9780122384523/a-mathematical-introduction-to-logic

Overview: Enderton presents a classic and systematic introduction to mathematical logic. The author covers propositional logic, first-order logic, truth, definability, undecidability, and model theory. The book supplies the essential logical tools used to analyze formal deductive systems.

Relevance to QBD: This logic reference is necessary for the epistemological foundations laid in Chapter 1. To support our coherentist approach to the selection of axioms, we must analyze the limits of deductive systems. Enderton's formal logic tools are required to demonstrate the inherent unprovability of axiomatic bases, establishing a sound logical starting point for our physical model.

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26. Erdos, P., & Renyi, A. (1960).

"On the evolution of random graphs" * **Link:** https://users.renyi.hu/~p_erdos/1960-10.pdf

Overview: Erdos and Renyi present the foundational paper on the evolution of random graphs, introducing the classical probabilistic model where edges are added stochastically. They prove the existence of sharp phase transitions, specifically the sudden appearance of a unique giant component as the average vertex degree exceeds one.

Relevance to QBD: This seminal work is the foundation for the geometrogenesis proofs in Chapter 11. We model the emergence of physical space as a phase transition in a random causal network. Erdos and Renyi's results supply the basis for this phase transition, showing that the vacuum graph stochastically transitions from a disjointed state to a unified, highly connected spacetime manifold.

---

27. Fuchs, C. A. (2010).

"QBism, The Perimeter of Quantum Bayesianism" * **Link:** https://arxiv.org/abs/1003.5209

Overview: Fuchs introduces QBism, an interpretation of quantum mechanics that combines quantum information theory with personalist Bayesian probability. The author argues that quantum states do not represent objective physical entities, but rather the personal probabilities and expectations of observers who interact with the physical world.

Relevance to QBD: QBism provides the epistemological backing for how quantum measurement is modeled in Chapter 4. In QBD, the update operator represents an active observation event that resolves relational quantum possibilities into concrete causal history. Fuchs's perspective warrants our treatment of these measurement events not as external perturbations, but as the fundamental relational update cycles of the universe.

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28. Gambini, R., Porto, R. A., Torterolo, S., & Pullin, J. (2009).

"Conditional probabilities with Dirac observables and the problem of time in quantum gravity" - *Physical Review D*, 79(4), 041501(R) * **Link:** https://doi.org/10.1103/PhysRevD.79.041501 (arXiv: https://arxiv.org/abs/0809.4235)

Overview: Gambini and his co-authors present a careful application of the Page-Wootters mechanism to canonical quantum gravity. They show that relational time can be successfully constructed within a fully stationary quantum state of the universe by computing conditional probabilities between the states of a physical clock and a target system.

Relevance to QBD: The Page-Wootters mechanism is the conceptual precursor to the relational time formulation developed in Chapter 14. In QBD, the universe is described by a global stationary state where physical time emerges from the entanglement between the clock graph and the system graph. The relational validity of this clock mechanism is confirmed here, demonstrating that time emerges naturally without a background coordinate system.

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29. Gilbarg, D., & Trudinger, N. S. (2001).

"Elliptic Partial Differential Equations of Second Order" - *Springer* * **Link:** https://link.springer.com/book/10.1007/978-3-642-61798-0

Overview: Gilbarg and Trudinger present a definitive and thorough treatment of classical elliptic partial differential equations. They cover maximum principles, Sobolev spaces, Schauder estimates, and existence theorems, supplying the standard analytical tools used to analyze smooth geometric operators.

Relevance to QBD: This reference is necessary for the discrete field equations formulated in Chapter 13. To prove that the discrete Einstein field equations converge to the classical continuous equations, we must analyze the properties of elliptic operators on the manifold. Gilbarg and Trudinger's analytical tools bound the convergence errors of these operators, ensuring a mathematically consistent limit.

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30. Gillespie, D. T. (1977).

"Exact stochastic simulation of coupled chemical reactions" - *The Journal of Physical Chemistry*, 81(25), 2340-2361 * **Link:** https://pubs.acs.org/doi/10.1021/j100540a008

Overview: Gillespie develops the Stochastic Simulation Algorithm (SSA), a precise numerical method used to simulate the time evolution of coupled chemical reactions in a well-mixed volume. By integrating the reaction probabilities stochastically, the algorithm provides exact realizations of the master equation, capturing the discrete fluctuations that are ignored by deterministic rate equations.

Relevance to QBD: The Gillespie algorithm is the numerical foundation for the stochastic update simulations conducted in Chapter 4. We model the application of the rewrite rules as a set of coupled stochastic reactions where the graph vertices behave as reactants. Gillespie's method anchors the exact stochastic simulation used to validate that the graph evolves toward a stable macroscopic vacuum.

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31. Godel, K. (1931).

"On Formally Undecidable Propositions of Principia Mathematica and Related Systems" * **Link:** https://homepages.uc.edu/~martinj/History_of_Logic/Godel/Godel%20%E2%80%93%20On%20Formally%20Undecidable%20Propositions%20of%20Principia%20Mathematica%201931.pdf

Overview: Godel presents the monumental proof of the incompleteness theorems, demonstrating that any consistent formal axiomatic system capable of doing arithmetic contains propositions that are true but undecidable within the system. This work reveals the inherent limits of axiomatic formalization, transforming the foundations of mathematics.

Relevance to QBD: Godel's incompleteness theorems provide the logical motivation for the epistemological stance adopted in Chapter 1. Because no formal system can prove its own consistency, a physical model cannot claim absolute security. Godel's results warrant our shift from foundationalist justification to a pragmatic, coherentist epistemology, confirming that our axioms must be judged by their macroscopic consequences rather than claims of self-evidence.

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32. Gorard, J. (2020).

"Some Relativistic and Gravitational Properties of the Wolfram Model" - *Complex Systems*, 29(2), 599-654 * **Link:** https://doi.org/10.25088/ComplexSystems.29.2.599

Overview: Gorard analyzes the mathematical properties of multiway causal graphs and rewrite systems, establishing connections between causal invariance, discrete differential geometry, and the Einstein field equations. In particular, he investigates the convergence of discrete causal graphs to continuous pseudo-Riemannian spacetimes and explores the spectral properties of graph rewrite generators.

Relevance to QBD: Gorard's analysis establishes the critical adjoint kernel theorem formalized in Chapter 12 and Chapter 13: on weakly connected discrete state spaces, the kernel of the adjoint generator is strictly 1-dimensional ($\ker(\mathcal{L}^\dagger) = \text{span}\{\mathbf{1}\}$). This rules out non-trivial tensor collision invariants at the discrete level and explains why naive discrete moment expansions cannot close a tensor hydrodynamic hierarchy. In QBD, this obstruction is resolved by deriving the field equations via the scalar variational action on the category of histories and modular entanglement equilibrium.

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33. Gottesman, D. (1997).

"Stabilizer Codes and Quantum Error Correction" * **Link:** https://arxiv.org/abs/quant-ph/9705052

Overview: Gottesman introduces the stabilizer formalism, a powerful algebraic structure that simplifies the design and analysis of quantum error-correcting codes. By representing quantum codes in terms of their stabilizer groups, he provides the essential tools used to construct fault-tolerant quantum gates and correct arbitrary errors.

Relevance to QBD: The stabilizer formalism is the leading tool used to protect the topological graph structures in QBD. In Chapter 10, we utilize Gottesman's formalism to define stabilizer operators on the vertices and edges of our tripartite braid configurations. This ensures that the logical information remains protected from local vacuum fluctuations, demonstrating that topological qubits are stable.

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34. Gukov, S., Takayanagi, T., & Toumbas, N. (2004).

"Flux backgrounds in 2D string theory" * **Link:** https://arxiv.org/abs/hep-th/0312208

Overview: Gukov, Takayanagi, and Toumbas analyze the properties of two-dimensional string theory in the presence of background fluxes. They focus on how these backgrounds affect the compactification geometry, demonstrating that non-trivial topological configurations generate stable, localized energy density within the compactified space.

Relevance to QBD: This string-theoretic analysis provides the conceptual backing for the compactification models developed in Chapter 17. In QBD, the compactification of the causal graph along a toroidal boundary is shown to yield localized topological invariants. Gukov's results supply the physical precedent for how background fluxes stabilize these compactified geometries, ensuring the stability of emergent physical coordinates.

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35. Harlow, D. (2016).

"Jerusalem Lectures on Black Holes and Quantum Information" * **Link:** https://arxiv.org/abs/1409.1231

Overview: Harlow presents a comprehensive set of lectures on the role of quantum information theory in black hole physics, focusing on the black hole information paradox and the holographic principle. He

demonstrates that AdS/CFT bulk reconstruction can be modeled as a quantum error-correcting code with systematic rigor.

Relevance to QBD: Harlow's lectures provide the chief inspiration for the holographic bulk/boundary correspondence proved in Chapter 16. In QBD, the interior geometry of the causal graph behaves as a protected logical bulk that is mapped to a boundary screen. Harlow's quantum informational treatment corroborates this code model, showing that spacetime geometry is a holographic error-correcting code.

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36. Hawking, S. W., & Ellis, G. F. R. (1973).

"The Large Scale Structure of Space-Time" * **Link:** https://doi.org/10.1017/CBO9780511524646

Overview: Hawking and Ellis present the definitive monograph on the large-scale global structure of spacetime in general relativity. They develop the apparatus of differential geometry and global analysis to prove the classic singularity theorems, demonstrating that singularities are inevitable occurrences in classical general relativity.

Relevance to QBD: This seminal textbook is the direct reference for the classical Lorentzian geometry targeted in Chapter 11. To prove that the discrete causal graph converges to a physical spacetime, we must compare the discrete shortest-path metric to the continuous Lorentzian metric. Hawking and Ellis's treatment ensures that our definitions of causal order and light-cone structure align with established general relativity.

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37. Hossenfelder, S. (2011).

"Comments on and Comments on Comments on Verlinde's Entropic Gravity" - *Physica Scripta*, 2011(T140), 014067 * **Link:** https://arxiv.org/abs/1003.1015

Overview: Hossenfelder critiques entropic gravity frameworks that define gravitational attraction as an entropic force $\vec{F} = T\nabla S$. She proves that such entropic forces are fundamentally dissipative: in time-dependent backgrounds or closed periodic orbits, the work integral $\oint \vec{F} \cdot d\vec{r}$ fails to vanish, leading to non-conservative energy loss and catastrophic orbital decay for planetary and celestial systems.

Relevance to QBD: Hossenfelder's critique establishes an essential benchmark for the viability of emergent gravity. In Chapter 13 and Chapter 14, QBD proves its complete immunity to this pathology: gravitational interactions are not mediated by an irreversible thermodynamic gradient across a thermal bath, but arise strictly from a Stationary Action Principle ($\delta\mathcal{S} = 0$) on the causal network. Because homeostatic equilibrium satisfies exact local detailed balance, closed-loop orbital dissipation vanishes identically ($\oint \vec{F} \cdot d\vec{r} = 0.000000\,\mathrm{J}$), preserving conservative Hamiltonian mechanics.

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38. Jacobson, T. (1995).

"Thermodynamics of Spacetime: The Einstein Equation of State" * **Link:** https://arxiv.org/abs/gr-qc/9504004

Overview: Jacobson derives the Einstein field equations of general relativity directly from thermodynamic relations, demonstrating that gravity can be interpreted as an emergent equation of state. By applying the Clausius relation ($dS = dQ/T$) to a local causal horizon, he proves that the Einstein equation is a necessary consequence of horizon thermodynamics.

Relevance to QBD: Jacobson's emergent gravity derivation is a key physical pillar for the geometrogenesis proofs in Chapter 13. In QBD, the discrete field equations are shown to emerge from the thermodynamic equilibrium of the vacuum graph. Jacobson's results underpin our interpretation of gravity as a macroscopic equation of state, confirming that the curvature of spacetime arises from localized information entropy.

39. Jacobson, T. (2016).

"Entanglement Equilibrium and the Einstein Equation Update" - *Physical Review Letters*, 116(20), 201101 * **Link:** https://doi.org/10.1103/PhysRevLett.116.201101

Overview: Jacobson updates and fundamentally deepens his 1995 thermodynamic derivation of general relativity by replacing the phenomenological Clausius heat relation ($\delta Q = TdS$) with the First Law of Entanglement Entropy ($\delta S_{\text{ent}} = \delta\langle K\rangle$) evaluated across small geodesic spheres. He demonstrates that the requirement that the vacuum entanglement entropy is maximal at fixed volume directly implies the full nonlinear Einstein field equations with a cosmological constant.

Relevance to QBD: Jacobson's (2016) modular entanglement equilibrium provides the primary theoretical architecture for the continuous field equation derivation in Chapter 14. By anchoring the derivation in the Von Neumann entanglement entropy of boundary cut-sets and the modular Hamiltonian of matter defect fluxes, QBD completely bypasses the need for an external thermal heat bath or horizon temperature, anchoring gravity purely in the quantum information theory of the causal graph.

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40. Janson, S. (1987).

"Poisson approximation for large cycles in random graphs" * **Link:** http://stat.wharton.upenn.edu/~steele/Courses/531/531Resoureces/Janson1987PoissonProcess.pdf

Overview: Janson develops a careful probabilistic treatment to study the distribution of large cycles in random graphs. He utilizes Poisson approximation methods and correlation inequality techniques to prove that the count of disjoint cycles converges to a Poisson process in the sparse limit, establishing precise limits on local correlation.

Relevance to QBD: This probabilistic toolset is indispensable for the vacuum stability proofs in Chapter 5. We must show that the local cycles in our vacuum graph do not cluster or trigger a runaway collapse. Janson's bounds confirm that the local cycle density remains stable, supporting the claim that the vacuum background spacetime remains flat.

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41. Jaynes, E. T. (1957).

"Information Theory and Statistical Mechanics" - *Physical Review*, 106(4), 620-630 * **Link:** https://doi.org/10.1103/PhysRev.106.620

Overview: Jaynes formulates statistical mechanics as a form of statistical inference based on Shannon's information theory. He demonstrates that thermodynamic entropy is an informational measure of missing microscopic information, and that the Boltzmann-Gibbs canonical distribution maximizes information entropy subject to expectation constraints without requiring mechanical ergodic hypotheses.

Relevance to QBD: Jaynes's Maximum Entropy Principle provides the foundational justification for the microscopic update engine in Chapter 4 and Chapter 5. In QBD, probability is fundamental while temperature is not: the microscopic boolean edge space $\{0, 1\}$ is governed by Jaynes MaxEnt subject to topological cycle constraints. The critical parameter $\beta_c = \ln 2$ is established strictly as an informational base conversion modulus ($1\,\text{bit} = \ln 2\,\text{nats}$) rather than an inverse thermodynamic temperature, eliminating fictitious vacuum heat baths.

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42. Jones, V. F. R. (1985).

"A polynomial invariant for knots via a von Neumann algebra" * **Link:** https://www.ams.org/bull/1985-12-01/S0273-0979-1985-15304-2/

Overview: Jones introduces the Jones polynomial, a revolutionary invariant for knots and links that is constructed using trace formulas on von Neumann algebras. This work bridges the gap between low-dimensional topology and statistical mechanics, laying the foundation for modern knot theory and topological quantum field theory.

Relevance to QBD: The Jones polynomial is the direct topological invariant used to protect the particle braids in Chapter 6. To prove that the tripartite braid cannot be untied by local rewrite operations, we show that its Jones polynomial remains invariant under local moves. Jones's algebraic construction anchors this topological lock, demonstrating that fermions are topologically protected defects.

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43. Jost, J., & Liu, S. (2016).

"Ollivier's Ricci curvature, local clustering and curvature-dimension inequalities on graphs" * **Link:** https://arxiv.org/abs/1103.4037

Overview: Jost and Liu analyze Ollivier's definition of Ricci curvature on discrete graphs, proving that it correlates with the graph's local clustering coefficient. They derive key curvature-dimension inequalities that govern how diffusion processes behave on curved discrete networks, establishing a rigorous connection to continuous differential geometry.

Relevance to QBD: This discrete curvature analysis is pivotal for the geometrogenesis proofs in Chapter 13. In QBD, the discrete field equations are formulated by defining Ollivier-Ricci curvature along the edges of the causal graph. Jost and his co-authors' calculus provides the tools to calculate this curvature, confirming that the graph's connectivity relates directly to physical spacetime curvature.

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44. Kitaev, A. Y. (2003).

"Fault-tolerant quantum computation by anyons" * **Link:** https://arxiv.org/abs/quant-ph/9707021

Overview: Kitaev introduces the toric code, a revolutionary quantum error-correcting code defined on a two-dimensional lattice. He demonstrates that storing quantum information in the global topological properties of the lattice protects it from local environment noise, proving that fault-tolerant quantum computation can be achieved using topological anyons.

Relevance to QBD: Kitaev's toric code is the foremost conceptual model for the stabilizer-protected graph structures developed in Chapter 10. We leverage his insights to prove that our tripartite braid configurations are stable under local rewrite noise. Kitaev's treatment establishes the topological quantum computing structure used to model our particles as stable qubits in the causal graph.

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45. Kobakhidze, A. (2011).

"Once More on the Entropic Origin of Gravity" - *Physical Review D*, 84(4), 044031 * **Link:** https://doi.org/10.1103/PhysRevD.84.044031

Overview: Kobakhidze presents a decisive critique of entropic gravity theories, pointing out that if gravity arises from an ambient thermal bath at the Unruh temperature, quantum particles (such as ultracold neutrons in Earth's gravitational field) must experience severe thermal decoherence. He demonstrates that experiments measuring discrete gravitational bound states (e.g., Nesvizhevsky et al. 2002) decisively rule out any gravity model involving thermal environmental decoherence on observable timescales.

Relevance to QBD: Kobakhidze's critique serves as an exacting stress-test for Quantum Braid Dynamics in Chapter 13 and Chapter 14. QBD demonstrates complete immunity to this critique: the vacuum is an informational quantum ground state ($\Delta U = 0$) rather than a thermal gas. Discreteness fluctuations are suppressed by the square of the Planck-to-wavepacket ratio $(\ell_0/z_1)^2 \approx 1.39 \times 10^{-60}$, yielding a quantum coherence lifetime $\tau_{\text{QBD}} > 10^{59}$ s and fully preserving unitary quantum mechanics.

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46. Lamport, L. (1978).

"Time, clocks, and the ordering of events in a distributed system" * **Link:** https://doi.org/10.1145/359545.359563

Overview: Lamport introduces the seminal concept of logical clocks to establish a partial ordering of events in distributed systems without relying on synchronized physical clocks. By defining a relational happened-before relation based on message passing and event sequencing, he shows how independent nodes can construct a consistent global ordering of events. This paper laid the groundwork for modern distributed systems by demonstrating that logical ordering is more fundamental than physical time.

Relevance to QBD: Lamport's logical clock formalism is the starting point for the dual-time architecture developed in Chapter 14. In QBD, the local update events are partially ordered on the causal graph, representing the discrete happened-before relations. We leverage Lamport's logical clocks to construct a globally consistent historical timeline from these distributed updates, showing that emergent physical time arises naturally from discrete relational ordering.

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47. Landauer, R. (1991).

"Information is Physical" * **Link:** https://doi.org/10.1063/1.881299

Overview: Landauer argues that information cannot exist independently of a physical representation, meaning that processing and storing information are governed by physical laws. He reviews Landauer's principle, which dictates that any logically irreversible operation, such as the erasure of a bit, must dissipate a minimum amount of heat into the environment. This work established a deep physical connection between information theory, computation, and thermodynamics.

Relevance to QBD: This physical principle is foundational for the dynamical rewrite engine formulated in Chapter 4. In QBD, the deletion of edges during the update cycles constitutes a logically irreversible erasure of topological information. Landauer's principle establishes the physical necessity of localized heat dissipation during these deletions, confirming that the energetic cost of quantum gravity updates is fundamentally linked to information thermodynamics.

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48. Leibniz-Clarke Correspondence (1715-1716).

"The Leibniz-Clarke Correspondence" - *Ariew Edition* * **Link:** https://personal.lse.ac.uk/robert49/teaching/ph103/pdf/Ariew_1715LeibnizClarkeCorrespondence.pdf

Overview: The Leibniz-Clarke correspondence records a classic philosophical debate on the nature of space and time. Leibniz advocates for a relational view, arguing that space and time are systems of relations between coexisting objects. Clarke, defending Newton's view, argues that space and time are absolute, infinite containers within which physical objects are placed.

Relevance to QBD: This historical debate provides the conceptual framing for the relational substrate defined in Chapter 1. QBD rejects Newtonian absolute space in favor of a relational structure where space and time emerge solely from the connectivity of the causal graph. This correspondence frames our alignment

with Leibniz's relational view, demonstrating that our graph-theoretic model is a realization of his relational philosophy.

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49. Liggett, T. M. (1999).

"Stochastic Interacting Systems: Contact, Voter and Exclusion Processes" - *Springer-Verlag Berlin Heidelberg* * **Link:** https://doi.org/10.1007/978-3-662-03990-8

Overview: Liggett provides a comprehensive treatise on non-equilibrium particle systems, establishing rigorous theorems for contact processes, survival probabilities, and invariant measures on graphs and regular trees.

Relevance to QBD: Liggett's rigorous results on survival regimes and quasi-stationary distributions on Bethe lattices justify the persistence of local cycle density clusters in Chapter 5, demonstrating why discrete graph contact dynamics depart fundamentally from homogeneous mean-field approximations.

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50. Maldacena, J. M. (1998).

"The Large N Limit of Superconformal Field Theories and Supergravity" * **Link:** https://arxiv.org/abs/hep-th/9711200

Overview: Maldacena introduces the Anti-de Sitter / Conformal Field Theory (AdS/CFT) correspondence, proposing a duality between a gravity theory in the bulk of a spacetime and a gauge theory on its boundary. This holographic duality proves that continuous gravitational degrees of freedom can be completely mapped to lower-dimensional, non-gravitational quantum field theories.

Relevance to QBD: This seminal duality provides the central conceptual paradigm for the holographic screens developed in Chapter 16. In QBD, the interior causal graph represents the gravitational bulk, which is mapped to a discrete boundary screen through code mappings. Maldacena's correspondence grounds the theoretical precedent for our discrete holographic mapping, demonstrating that our graph-theoretic bulk arises from a boundary code.

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51. Marker, D. (2002).

"Model Theory: An Introduction" * **Link:** https://link.springer.com/book/10.1007/b98860

Overview: Marker presents a comprehensive introduction to model theory, focusing on the relationship between formal mathematical languages and their interpretations. He covers key topics such as compactness, completeness, quantifier elimination, and the properties of first-order theories, supplying the formal logic tools used to analyze structures.

Relevance to QBD: This reference is necessary for the logical and model-theoretic analyses conducted in Chapter 1. To ensure that the axioms of QBD are formally sound, we evaluate their models on our discrete graph substrate. Marker's textbook provides the foundations for these evaluations, helping us verify that our relational update rules represent a consistent first-order theory.

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52. Mousa, M., Jamadagni, A., et al. (2025).

"Pauli Stabilizer Models for Gapped Boundaries of Twisted Quantum Doubles and Applications to Composite Dimensional Codes" * **Link:** https://arxiv.org/abs/2508.19245

Overview: Mousa and his collaborators construct Pauli stabilizer models to describe the gapped boundaries of twisted quantum double models. They show how these boundary stabilizers can be utilized to construct

composite dimensional error-correcting codes, providing robust protection against local noise in topologically ordered systems.

Relevance to QBD: This recent stabilizer model is vital for the boundary protection theories developed in Chapter 10. In QBD, the tripartite braid configurations must remain stable near the boundaries of the causal graph. Mousa's algebraic machinery supports the construction of stable boundary stabilizers on the graph, confirming that our topological qubits remain protected even in the presence of edge boundaries.

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53. Ollivier, Y. (2009).

"Ricci curvature of Markov chains on metric spaces" * **Link:** https://arxiv.org/pdf/math/0701886

Overview: Ollivier develops a robust approach to define Ricci curvature on arbitrary metric spaces using transport distances between probability measures. He shows that this definition, known as Ollivier-Ricci curvature, captures the geometric properties of continuous Riemannian manifolds while remaining fully applicable to discrete networks.

Relevance to QBD: Ollivier's metric curvature is the direct tool used to formulate the discrete field equations in Chapter 13. By calculating the transport distance between localized random walks on our causal graph, we define the Ollivier-Ricci curvature along each edge. Ollivier's calculus provides the formal apparatus used to prove that this discrete curvature converges to classical Ricci curvature.

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54. Otto, F., Mansuroglu, R., Schuch, N., Guhne, O., & Sahlmann, H. (2025).

"Hyperinvariant Spin Network States: An AdS/CFT Model from First Principles" * **Link:** https://arxiv.org/abs/2510.06602

Overview: Otto and his co-authors present a first-principles derivation of hyperinvariant spin network states in discrete gravity, establishing an AdS/CFT model that operates on hyperbolic geometries. They prove that these spin networks exhibit robust entanglement properties that match the holographic predictions of continuous gravity.

Relevance to QBD: This work provides decisive validation for the spin-network formulations developed in Chapter 13. QBD models the causal graph as a network of spin-like connections where spatial geometry is reconstructed via edge entanglement. The algebraic connection between our discrete graph states and the holographic spin networks of loop quantum gravity is traced through this paper.

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55. Padmanabhan, T. (2009).

"Thermodynamical Aspects of Gravity: New Insights" * **Link:** https://arxiv.org/abs/0911.5004

Overview: Padmanabhan reviews the thermodynamic description of gravity, presenting extensive evidence that gravity is not a fundamental interaction but rather an emergent thermodynamic phenomenon. He demonstrates that the field equations can be written as a local thermodynamic identity on causal horizons, linking geometry directly to entropy.

Relevance to QBD: Padmanabhan's thermodynamic analysis is a central conceptual foundation for the emergent gravity proofs in Chapter 13. In QBD, spatial curvature emerges from the thermodynamic equilibrium of the vacuum graph. His review provides the physical motivation for treating general relativity as a macroscopic equation of state, linking discrete updates to thermodynamic entropy.

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56. Page, D. N. (1993).

"Information in Black Hole Radiation" * **Link:** https://arxiv.org/abs/hep-th/9306083

Overview: Page analyzes the entanglement entropy of a quantum system undergoing unitary evaporation, deriving what is now known as the Page curve. He proves that if the evaporation process is unitary, the entanglement entropy of the radiation must first rise and then return to zero, establishing a key benchmark for resolving the information paradox.

Relevance to QBD: The Page curve is a key physical benchmark used to verify the unitarity of the rewrite engine in Chapter 16. In QBD, the evaporation of topological graph defects is modeled as a unitary process on the causal network. Page's analysis provides the model used to confirm that our discrete update rules successfully preserve quantum information, preventing information loss.

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57. Page, D. N., & Wootters, W. K. (1983).

"Evolution without evolution: Dynamics described by stationary observables" * **Link:** https://journals.aps.org/prd/abstract/10.1103/PhysRevD.27.2885

Overview: Page and Wootters formulate a relational interpretation of quantum mechanics where time arises from the entanglement between a subsystem acting as a clock and the rest of the universe. In this formulation, the global state of the universe is completely stationary, and dynamical evolution emerges relational when conditioning on the clock's state.

Relevance to QBD: This relational time approach is the core architecture used to solve the problem of time in Chapter 14. In QBD, the global quantum state on the causal graph is stationary, and dynamical physical time emerges relationally from the entanglement between the clock and the substrate. Page and Wootters's construction grounds the relational validity of our dual-time mechanism.

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58. Palmigiano, A., & Sadrzadeh, M. (Eds.). (2023).

"Samson Abramsky on Logic and Structure in Computer Science and Beyond" * **Link:** https://link.springer.com/book/10.1007/978-3-031-24117-8

Overview: Palmigiano and Sadrzadeh edit a comprehensive festschrift honoring Samson Abramsky, compiling chapters that explore the structural connections between logic, category theory, and quantum mechanics. The text highlights Abramsky's contributions to categorical quantum mechanics, game semantics, and structural models of contextuality.

Relevance to QBD: This volume is the direct reference for the categorical quantum mechanics models used in Chapter 1. We leverage Abramsky's structural formulations of quantum processes to represent the update engine as a functor between categories. The category-theoretic foundations required to model relational quantum processes on our causal graph are drawn from this collection.

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59. Pastawski, F., Yoshida, B., Harlow, D., & Preskill, J. (2015).

"Holographic quantum error-correcting codes: Toy models for the bulk/boundary correspondence" * **Link:** https://arxiv.org/abs/1503.06237

Overview: Pastawski and his co-authors introduce the HaPPY code, a toy model for AdS/CFT bulk reconstruction constructed using pentagon tensor networks. They prove that the bulk geometry is robustly mapped to the boundary through a holographic quantum error-correcting code, providing a concrete realization of bulk protection from boundary errors.

Relevance to QBD: The HaPPY code is the direct template for the holographic screen mechanisms developed in Chapter 16. In QBD, we model the interior of the causal graph as a logical bulk protected by boundary stabilizers. The HaPPY tensor-network structure maps bulk coordinates to boundary screens, showing that space is a holographic error-correcting code.

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60. Pemantle, R. (1992).

"The contact process on trees" - *Annals of Probability*, 20(4), 2089-2116. * **Link:** https://doi.org/10.1214/aop/1176989541

Overview: Pemantle proves that the contact process on infinite regular trees exhibits two distinct phase transitions $\lambda_{c1} < \lambda_{c2}$, where λ_{c1} marks the onset of local survival (non-zero probability of surviving near the origin indefinitely) and λ_{c2} marks the onset of global survival (survival with positive probability throughout the tree).

Relevance to QBD: This two-threshold theorem provides the rigorous mathematical foundation for resolving the mean-field extinction paradox in Chapter 5. It proves that localized topological defects (solitons) can persist in a non-equilibrium quasi-stationary distribution on branching pre-geometric substrates without triggering an unphysical runaway percolation cascade.

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61. Penington, G. (2019).

"Entanglement Wedge Reconstruction and the Information Paradox" * **Link:** https://arxiv.org/abs/1905.08255

Overview: Penington proves that the entanglement entropy of Hawking radiation follows the Page curve by incorporating quantum extremal surfaces into the calculation. He demonstrates that the holographic entanglement wedge of the black hole interior shifts dynamically, showing that bulk information is reconstructed from boundary radiation.

Relevance to QBD: This holographic reconstruction is central to the black hole simulation audits conducted in Chapter 16. In QBD, the evaporation of localized graph singularities is analyzed using discrete quantum extremal surfaces. Penington's holographic apparatus shows that the interior bulk graph is unitarily reconstructed from boundary screen updates, preserving information.

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62. Rodrigues, F. L. S., & Lutz, E. (2025).

"Far-from-equilibrium thermodynamics of non-Abelian thermal states" * **Link:** https://arxiv.org/abs/2510.04788

Overview: Rodrigues and Lutz develop a thermodynamic treatment of non-Abelian thermal states in systems operating far from equilibrium. They derive key fluctuation relations and entropy production bounds that govern how non-Abelian gauge configurations thermalize and dissipate energy under non-equilibrium conditions.

Relevance to QBD: This non-equilibrium thermodynamic analysis is indispensable for the non-Abelian gauge models developed in Chapter 8. In QBD, the tripartite braids operate as non-Abelian states that undergo far-from-equilibrium updates during the transition cycles. The thermodynamic bounds derived here are used to analyze the stability of these non-Abelian states against runaway vacuum dissipation.

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63. Rovelli, C. (1996).

"Relational Quantum Mechanics" * **Link:** https://arxiv.org/abs/quant-ph/9609002

Overview: Rovelli introduces Relational Quantum Mechanics (RQM), postulating that quantum states do not represent absolute properties of physical systems but rather relational information between systems. He argues that physical systems are completely defined by the relations they establish with other systems, eliminating the need for an absolute observer.

Relevance to QBD: RQM is the central epistemological foundation for the update dynamics formulated in Chapter 4. In QBD, the state of the causal graph is entirely relational, where vertices possess states only relative to neighboring connections. Rovelli's relational model provides the physical motivation for this approach, showing that quantum measurement is a fundamental relational update event on the graph.

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64. Rovelli, C., & Smolin, L. (1990).

"Loop space representation of quantum general relativity" * **Link:** https://doi.org/10.1016/0550-3213(90)90019-A

Overview: Rovelli and Smolin construct the loop space representation of quantum general relativity, formulating gravity in terms of loops and connections. They prove that the spatial geometry is quantized in terms of discrete spin network states, establishing a non-perturbative structure for what would become Loop Quantum Gravity.

Relevance to QBD: This loop space representation is the foremost conceptual template for the spatial coordinates formulated in Chapter 13. In QBD, spatial geometry is encoded in connection-like loops along the edges of the causal graph. This classic work traces the algebraic connection between our discrete, edge-based connections and the spin networks of loop quantum gravity.

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65. Ryu, S., & Takayanagi, T. (2006).

"Holographic Derivation of Entanglement Entropy from AdS/CFT" * **Link:** https://arxiv.org/abs/hep-th/0603001

Overview: Ryu and Takayanagi propose a holographic formula to calculate the entanglement entropy of a boundary conformal field theory CFT using the area of a minimal surface in the dual AdS bulk spacetime. This formula, known as the Ryu-Takayanagi RT formula, establishes a direct geometric link between spatial area and quantum entanglement.

Relevance to QBD: The Ryu-Takayanagi formula is the direct tool used to calculate bulk geometry from boundary entanglement in Chapter 16. In QBD, the spatial area of emergent regions is shown to match the boundary entanglement entropy calculated along minimal cuts of the graph. This reference validates the geometric entanglement area proofs.

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66. Sachs, H. (1962).

"Uber selbstkomplementare Graphen" - *Publicationes Mathematicae Debrecen*, 9, 270-288 * **Link:** https://scispace.com/pdf/uber-selbstkomplementare-graphen-2cpuwz9n.pdf

Overview: Sachs presents the foundational work on self-complementary graphs, which are graphs that are isomorphic to their own complement. He derives key algebraic properties and structural constraints that govern the distribution of edges in these graphs, establishing precise bounds on their cycle structure.

Relevance to QBD: This reference is necessary for the tripartite braid audits conducted in Chapter 6. We model the stable particle braids using self-complementary topological configurations. Sachs's structural

constraints confirm that these self-complementary configurations are protected from untying by local graph updates, supporting the stability of fermions.

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67. Sakharov, A. D. (1967).

"Vacuum Quantum Fluctuations in Curved Space and the Theory of Gravitation" - *Soviet Physics Doklady*, 12, 1040-1041 * **Link:** https://doi.org/10.1070/PU1991v034n05ABEH002498

Overview: Sakharov introduces the concept of "induced gravity," proposing that the gravitational field and the Einstein-Hilbert action are not fundamental dynamical fields, but rather represent the macroscopic elastic response of spacetime metric geometry induced by the quantum fluctuations of microscopic matter and gauge fields. In this framework, Newton's constant G measures the elasticity of the quantum vacuum.

Relevance to QBD: Sakharov's induced metric elasticity directly underpins the derivation of Newton's constant in Chapter 14. In QBD, the metric tensor and its curvature action emerge from integrating out the microscopic combinatorial rewrite transitions on the causal network. Newton's constant is derived as $G = \frac{c^3 \ell_0^2}{4\hbar\rho_3^*}$, representing the induced elastic modulus of the vacuum resisting the deformation of its homeostatic 3-cycle equilibrium density $\rho_3^* \approx 0.0370$.

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68. Sati, H., & Schreiber, U. (2025).

"The quantum monadology" * **Link:** https://ncatlab.org/schreiber/files/QuantumMonadology-250718.pdf

Overview: Sati and Schreiber formulate the quantum monadology, a categorical language that interprets quantum states and observers within a relational, category-theoretic context. Drawing inspiration from Leibnizian philosophy, they model the universe as a network of quantum monads that observe each other relationally. This categorical formalism establishes a precise language for describing how global quantum states can emerge from local, relational observations.

Relevance to QBD: This categorical formulation is indispensable for the relational model defined in Chapter 1. We adopt Sati and Schreiber's quantum monadology to formalize the interactions between local graph vertices as relational observations. Sati and Schreiber's category-theoretic tools show that global spacetime arises naturally from these localized, relational updates on the causal graph.

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69. Singer, A., & Wu, H.-T. (2013).

"Vector diffusion maps and the connection graph Laplacian" * **Link:** https://arxiv.org/abs/1102.0075

Overview: Singer and Wu introduce vector diffusion maps (VDM), a geometric approach that generalizes Laplacian eigenmaps to vector bundles on manifolds. They define the connection graph Laplacian, proving that its spectral properties recover both the underlying manifold's geometry and the gauge connection of the vector bundle, establishing a powerful tool for analyzing curved datasets.

Relevance to QBD: This connection graph Laplacian is the direct tool used to analyze the emergent gauge fields in Chapter 12. To show that our discrete graph connectivity yields continuous gauge fields, we must construct a vector bundle over the graph. Singer and Wu's spectral convergence proofs show how the eigenvectors of the connection Laplacian recover both physical coordinates and gauge connections.

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70. Sorkin, R. D. (2005).

"Causal sets: Discrete gravity" - *In Lectures on Quantum Gravity (pp. 305-327). Springer* * **Link:** https://arxiv.org/abs/gr-qc/0309009

Overview: Sorkin presents a comprehensive review of the causal set approach to quantum gravity, postulating that spacetime is fundamentally discrete and represented by a partially ordered set (poset) of events. He demonstrates that discrete causal relations are sufficient to recover the causal structure, topology, and volume of a continuous Lorentzian spacetime manifold.

Relevance to QBD: Sorkin's causal set model is a core physical pillar for the discrete causal substrate defined in Chapter 1. We adopt his insight that causality is fundamental and volume is discrete. However, we expand his poset setting by adding relational graph connectivity, which is necessary to support quantum states. Sorkin's work underpins the physical basis for our discrete spacetime model.

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71. Steinberg, M. et al. (2025).

"Universal Fault-Tolerant Logic with Heterogeneous Holographic Codes" * **Link:** https://arxiv.org/abs/2504.10386

Overview: Steinberg and his collaborators construct heterogeneous holographic codes, a class of tensor network error-correcting codes that support universal fault-tolerant logical gates. They prove that these codes provide robust bulk topological protection while maintaining universal logical operations, solving a major bottleneck in holographic quantum error correction.

Relevance to QBD: This holographic code model is indispensable for the logical gate simulations conducted in Chapter 10. In QBD, the tripartite braid configurations behave as bulk topological qubits that must support fault-tolerant logical operations under local graph updates. Steinberg's construction provides the theoretical setting used to model these operations as holographic gates, demonstrating that our particles are universal qubits.

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72. Uustalu, T., & Vene, V. (2008).

"Comonadic notions of computation" * **Link:** https://www.sciencedirect.com/science/article/pii/S1571066108003435

Overview: Uustalu and Vene formulate a comonadic approach to describe context-dependent computations in computer science. They demonstrate that while monads are effective at modeling computations that produce effects, comonads are the natural category-theoretic tool to model computations that rely on surrounding context or historical execution states.

Relevance to QBD: This comonadic structure is the direct tool used to formalize the local update rules in Chapter 2. Because our rewrite rules rely on the surrounding context of neighboring vertices and edges, they are modeled comonadically. This construction provides the category-theoretic foundations required to define these context-dependent updates, ensuring algebraic consistency.

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73. Vaccaro, J. A., & Barnett, S. M. (2011).

"Information Erasure Without an Energy Cost" - *Proceedings of the Royal Society A*, 467(2130), 1770-1778 * **Link:** https://doi.org/10.1098/rspa.2010.0577

Overview: Vaccaro and Barnett analyze the physical limits of Landauer's principle, demonstrating that the erasure of information does not fundamentally require an energy dissipation cost ($k_B T \ln 2$) if the system uti-

lizes an alternative conserved resource, such as spin angular momentum or structural history. They establish that information retention and processing can occur losslessly without continuous thermal dissipation.

Relevance to QBD: Vaccaro and Barnett's discovery provides the thermodynamic underpinning for the append-only Category of Histories (**Hist**) in Chapter 4. By retaining all past causal relations as an immutable structural ledger, the causal graph updates its topology with zero Landauer bit erasure ($\Delta S_{\text{erase}} = 0$). This guarantees that microscopic rewrite kinetics operate reversibly on the category level without heating the vacuum.

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74. van der Hoorn, P., & Stegehuis, C. (2020).

"Mean-field bounds for the k-core in random graphs" - *Electronic Journal of Probability*, 25 * **Link:** https://arxiv.org/abs/2008.01209

Overview: van der Hoorn and Stegehuis derive mean-field bounds for the size and structure of the k-core in random graphs, analyzing how these dense subgraphs emerge under random edge configurations. They prove that the k-core exhibits sharp threshold behavior, establishing precise bounds on graph connectivity and local dense clusters.

Relevance to QBD: This probabilistic analysis is necessary for the vacuum stability audits conducted in Chapter 5. To prove that the vacuum graph does not collapse into densely connected local clusters, we must bound the density of its k-cores. Van der Hoorn's bounds show that the k-cores of our vacuum graph remain sparse, ensuring the stability of flat spacetime.

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75. van Kampen, N. G. (1992).

"Stochastic Processes in Physics and Chemistry (2nd ed.)" - *North-Holland* * **Link:** https://books.google.com/books?id=N6II-6HlPxEC

Overview: van Kampen presents a classic and thorough textbook on stochastic processes in physical and chemical systems. He covers the master equation, Fokker-Planck equations, expansion methods, and the properties of stochastic transitions in systems operating near or far from thermodynamic equilibrium.

Relevance to QBD: This textbook is the direct reference for the stochastic master equations formulated in Chapter 4. In QBD, the local update rules are modeled as stochastic transitions whose probabilities are governed by a master equation. Van Kampen's analytical tools show that this master equation converges to a stable macroscopic vacuum, supporting our model.

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76. van Luxburg, U., Belkin, M., & Bousquet, O. (2008).

"Consistency of spectral clustering" * **Link:** http://misha.belkin-wang.org/papers/CLEM_08.pdf

Overview: van Luxburg, Belkin, and Bousquet prove the consistency of spectral clustering, demonstrating that the eigenvectors of the graph Laplacian converge systematically to the eigenfunctions of the continuous Laplace-Beltrami operator as the graph size approaches infinity. They establish explicit convergence bounds for different graph normalization schemes.

Relevance to QBD: This convergence proof is a main cornerstone for the dimensional reconstruction proofs in Chapter 12. To show that a discrete causal network converges to a continuous manifold, we rely on spectral clustering consistency. This paper provides the bounds needed to verify that the discrete eigenvectors recover continuous coordinates in the infinite-volume limit.

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77. Verlinde, E. (2011).

"On the Origin of Gravity and the Laws of Newton" * **Link:** https://arxiv.org/abs/1001.0785

Overview: Verlinde proposes that gravity is not a fundamental interaction but rather an entropic force arising from information changes on holographic screens. By combining Bekenstein's horizon thermodynamics with holographic principles, he derives Newton's laws and the Einstein field equations as emergent thermodynamic equations of state.

Relevance to QBD: Verlinde's entropic gravity is a central conceptual foundation for the discrete field equations formulated in Chapter 13. In QBD, the spatial curvature of the causal graph is shown to emerge from the entropic forces generated by local graph update fluxes. Verlinde's treatment supports our interpretation of gravity as an entropic force, showing that geometry is an emergent information phenomenon.

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78. Wang, W. (2024).

"Building holographic code from the boundary" * **Link:** https://arxiv.org/abs/2407.10271

Overview: Wang constructs a holographic error-correcting code directly from boundary representations, analyzing how bulk geometric structures are encoded in boundary entanglement states. He proves that the bulk/boundary mapping is robust under boundary perturbations, establishing a systematic template for reconstruction in quantum gravity.

Relevance to QBD: This boundary code construction provides vital validation for the holographic screen mechanisms developed in Chapter 16. In QBD, the bulk causal graph is mapped to a discrete boundary screen through code mappings. Wang's construction anchors the algebraic structure used to prove that bulk coordinates are unitarily reconstructed from boundary updates, confirming bulk stability.

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79. Wheeler, J. A. (1990).

"Information, physics, quantum: The search for links" * **Link:** https://philpapers.org/archive/WHEIPQ.pdf

Overview: Wheeler introduces the classic concept of 'it from bit', postulating that every physical entity, such as space, time, or particles, derives its existence from binary information exchanges. He argues that the physical world is fundamentally informational, meaning that physical laws are the logical rules governing how observers acquire and process bits.

Relevance to QBD: Wheeler's informational paradigm is the philosophical foundation for the entire QBD monograph. We adopt his 'it from bit' perspective by modeling the universe as a discrete network of relational information updates on a causal graph. This reference grounds our fundamental premise that physical space, time, and matter emerge solely from discrete, relational bits.

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80. Wilson, K. G. (1975).

"The renormalization group: Critical phenomena and the Kondo problem" * **Link:** https://journals.aps.org/rmp/abstract/10.1103/RevModPhys.47.773

Overview: Wilson presents the definitive formulation of the renormalization group, describing how the effective physical parameters of a quantum field theory shift as the system is viewed at different length scales. This work provides the tools required to analyze critical phase transitions and calculate continuous limits in quantum field theories.

Relevance to QBD: The renormalization group is the main tool used to calculate the continuum limit of the discrete field equations in Chapter 12. By grouping local graph updates into larger coarse-grained blocks,

we show that the discrete Laplacian converges to a continuous operator. Wilson's scaling theory underpins this convergence.

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81. Witten, E. (1989).

"Quantum Field Theory and the Jones Polynomial" * **Link:** https://projecteuclid.org/journals/communications-in-mathematical-physics/volume-121/issue-3/Quantum-field-theory-and-the-Jones-polynomial/cmp/1104178138.full

Overview: Witten constructs topological quantum field theory (TQFT) by showing that the Jones polynomial of a knot can be calculated as the partition function of a Chern-Simons gauge theory. This work bridges the gap between low-dimensional topology and quantum field theory, proving that topological invariants correspond to observable physical amplitudes.

Relevance to QBD: This seminal TQFT construction is the direct algebraic precursor to the particle braid formulations developed in Chapter 6. In QBD, the stable particle states are represented by braids whose physical amplitudes are governed by Chern-Simons topological invariants. Witten's results connect low-dimensional topology to our emergent quantum particles.

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82. Woess, W. (2000).

"Random Walks on Infinite Graphs and Groups" * **Link:** http://math.bme.hu/~gabor/oktatas/SztoM/Woess.RWonInfinGrGp.pdf

Overview: Woess presents a comprehensive and exacting study of random walks on infinite graphs, focusing on how the graph's algebraic and geometric structure governs diffusion processes. He covers transient and recurrent walks, spectral radii, and boundary behavior, establishing the standard tools for discrete diffusion.

Relevance to QBD: This reference is necessary for the discrete diffusion analyses conducted in Chapter 11. To prove that the causal graph converges to a continuous manifold, we analyze how localized random walks diffuse across the network. Woess's bounds confirm that this diffusion is stable and recovers the metric of continuous spacetime.

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83. Wolfram, S. (2002).

"A New Kind of Science" * **Link:** https://www.wolframscience.com/nks/

Overview: Wolfram presents an extensive study of cellular automata and other simple computational systems, arguing that complex structures in nature can emerge from simple, deterministic update rules. He proposes that the universe itself operates as a discrete cellular automaton, suggesting that computational rules are more fundamental than continuous physical equations.

Relevance to QBD: Wolfram's computational paradigm is a key conceptual precursor to the graph rewrite dynamics developed in Chapter 2. We adopt his core insight that discrete, localized update rules can generate complex, emergent physical structures. Citing this book frames our work within the history of discrete physical models, highlighting our use of network-based rewriting.

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84. Wolfram, S. (2020).

"A Project to Find the Fundamental Theory of Physics" * **Link:** https://writings.stephenwolfram.com/2020/04/finally-we-may-have-a-path-to-the-fundamental-theory-of-physics-and-its-beautiful/

Overview: Wolfram introduces the Wolfram Physics Project, proposing a formal computational framework where spacetime is represented by a discrete, hypergraph-like network that evolves under simple rewrite rules. He shows how general relativity, quantum mechanics, and special relativity can emerge as mathematical consequences of these discrete updates.

Relevance to QBD: This project is a direct conceptual and computational precursor to QBD. We build upon Wolfram's hypergraph framework by adding structured topological braids and quantum error correction to resolve the vacuum stability problem. Citing this proposal establishes the contemporary context of discrete network models, highlighting our integration of topological stability.

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85. Zurek, W. H. (2003).

"Decoherence, Einselection, and the Quantum Origins of the Classical" * **Link:** https://arxiv.org/abs/quant-ph/0105127

Overview: Zurek reviews the quantum decoherence program, explaining how interactions between a quantum system and its environment select a preferred set of stable classical states, a process known as einselection. He proves that decoherence naturally explains how classical objectivity emerges from the underlying quantum superposition states.

Relevance to QBD: Decoherence and einselection are the key physical mechanisms used to explain the emergence of classical causal history in Chapter 4. In QBD, the environment of the causal graph decoheres relational quantum states into stable, objective classical edges. Zurek's analysis provides the physical motivation for this emergence, bridging the quantum substrate and classical space.

?— title: "Appendix B: Python Simulations" sidebar_label: "B: Python Simulations" —

Full simulation code is available, under a strict license, at:

GitHub Repository: https://github.com/braiddynamics/qbd-portal

Return to Table of Contents

Nomenclature & Symbol Table

This table defines the standard notation used throughout the Quantum Braid Dynamics (QBD) monograph.

Symbol	Description	Context / First Used
$\hat{H}$	Hamiltonian constraint operator	Sec.1.1
$\mathfrak{S}$	A finite formal system	Sec.1.1.1
$\mathcal{A}$	The Axiomatic Basis (set of foundational postulates)	Sec.1.1.1
$\mathfrak{D}$	A Formal Deductive System tuple $(\mathcal{L}, \mathcal{A}, \mathcal{I})$	Sec.1.1.2
$\mathcal{L}$	The Formal Language (alphabet and grammar)	Sec.1.1.2
$\mathcal{I}$	The set of Rules of Inference	Sec.1.1.2
$\vdash$	Syntactic derivability (provability)	Sec.1.1.2
$\models$	Semantic entailment (truth)	Sec.1.1.2
Γ	A set of premises	Sec.1.1.2

Symbol	Description	Context / First Used
θ	A derived theorem	Sec.1.1.2
$\mathfrak{F}$	A consistent system capable of primitive recursive arithmetic	Sec.1.1.3
$\mathcal{G}$	The Godel sentence (true but unprovable)	Sec.1.1.3
$Con(\mathfrak{F})$	The consistency statement of system $\mathfrak{F}$	Sec.1.1.3
$\perp$	Logical contradiction	Sec.1.1.6
DAG	Directed Acyclic Graph (relational foundation of causal order)	Sec.1.2.1
V_A, V_B	Disjoint vertex partitions (Bipartite definition)	Sec.1.2.2
$\Pi(u, v)$	Directed Path (sequence of relational causality from vertex u to v)	Sec.1.2.3
π_{simple}	Simple Path (acyclic trajectory of causal influence)	Sec.1.2.4
$\Pi_2(u, v)$	2-Path (minimal unit of transitive mediation)	Sec.1.2.5
C	Cycle (general topological causal closure)	Sec.1.2.6
$(u, v), (v, u)$	2-Cycle (minimal unit of reciprocal causality and feedback)	Sec.1.2.7
(A, B, C)	3-Cycle (minimal closed loop enclosing elementary spatial area)	Sec.1.2.8
(t_L, t_{phys})	Dual Time Architecture (logical clock and emergent duration)	Sec.1.3.1
t_{phys}	Physical Time (emergent metric time)	Sec.1.3.2
π	Directed Path (sequence of relational causality)	Sec.1.3.2
π_2	2-Path (minimal unit of transitive mediation)	Sec.1.3.2
t_L	Global Logical Time (discrete iteration counter / coordinate clock)	Sec.1.3.3
$\mathbb{N}_0$	Set of non-negative integers (Domain of t_L)	Sec.1.3.3
U_{t_L}	Global state of the universe at step t_L	Sec.1.3.3
$\mathcal{U}$	Universal Evolution Operator	Sec.1.3.3
Ψ	Wavefunction of the universe	Sec.1.3.3.1
τ	Fictitious time parameter (Stochastic Quantization)	Sec.1.3.3.1
μ	Renormalization scale	Sec.1.3.3.1
$\hat{P}$	Permutation Operator (CAI interpretation)	Sec.1.3.3.2
$\mathcal{T}$	Unimodular Time variable	Sec.1.3.3.3
$\Lambda, \hat{\Lambda}$	Cosmological Constant (variable/operator)	Sec.1.3.3.3
U_0	The unique initial state	Sec.1.3.4

Symbol	Description	Context / First Used
$S(U)$	Information content/Entropy of state U	Sec.1.3.5
$\mathcal{O}(\cdot)$	Big O notation (asymptotic growth)	Sec.1.3.5
Ω_t	Set of admissible physical states at time t	Sec.1.3.5.1
b	Finite Branching factor	Sec.1.3.5.1
s_t	Surface area (active degrees of freedom)	Sec.1.3.5.1
δ_{holo}	Holographic scaling constant	Sec.1.3.5.1
T	Temporal Domain (Set of integers)	Sec.1.3.6.1
$\mathbb{Z}_{\leq 0}$	Set of non-positive integers (Infinite Past domain)	Sec.1.3.6.1
$\mathcal{H}_{\text{hist}}$	History sequence (set of operations)	Sec.1.3.6.1
μ	Mean of entropy production (Context: Statistics)	Sec.1.3.6.1
σ^2	Variance of entropy production	Sec.1.3.6.1
ΔI_k	Information bit contribution	Sec.1.3.6.1
Ω	Universal State Space (Set of all admissible graphs)	Sec.1.3.7.1
$\mathcal{P}_T$	Trajectory sequence (Context: Recurrence Proof)	Sec.1.3.7.1
$\prec$	Strict causal precedence	Sec.1.3.7.1
$\epsilon(op)$	Energy cost per operation	Sec.1.3.8.1
E_{total}	Total energy dissipated	Sec.1.3.8.1
k_B	Boltzmann constant	Sec.1.3.8.2
T	Temperature (Context: Thermodynamics)	Sec.1.3.8.2
$\hbar$	Reduced Planck constant	Sec.1.3.8.2
c	Speed of light	Sec.1.3.8.2
$G_{\mu\nu}$	Einstein Tensor	Sec.1.3.8.2
$T_{\mu\nu}$	Continuous stress-energy tensor	Sec.1.3.8.2
R_s	Schwarzschild Radius	Sec.1.3.8.2
R_n	The n-th Grim Reaper entity	Sec.1.3.9.1
G	A specific Causal Graph (V, E, H)	Sec.1.4.1
V	Set of Vertices (Abstract Events)	Sec.1.4.2
v, u, w	Individual vertices	Sec.1.4.2
E	Set of Directed Edges (Causal Relations)	Sec.1.4.3
e	Individual edge (u, v)	Sec.1.4.3
H	History Function (Local proper time mapping $E \to \mathbb{N}$)	Sec.1.4.4
$\text{In}(u)$	Set of incoming edges to vertex u	Sec.1.4.5
$\mathfrak{T}$	Elementary Task Space	Sec.1.5.1
ΔF	Change in Free Energy	Sec.1.5.1.1
$\mathfrak{T}_{add}$	Primitive Task: Edge Addition	Sec.1.5.2
$\mathfrak{T}_{del}$	Primitive Task: Edge Deletion	Sec.1.5.3
$P_{err}(R)$	Probability of paradox evasion at radius R	Sec.2.1

Symbol	Description	Context / First Used
(u, v)	The Directed Causal Link (Atomic relation $u \to v$)	Sec.2.1.1
E	The set of edges within the graph	Sec.2.1.1
$\implies$	Logical implication	Sec.2.2.1
$\forall$	Universal quantifier ("for all")	Sec.2.2.1
$\mathcal{T}_{self}$	Self-Loop Addition Operation	Sec.2.2.3
$\Omega(G)$	Cardinality of the set of Simple Paths	Sec.2.2.3
ΔS	Change in Entropy	Sec.2.2.3
k_B	Boltzmann Constant	Sec.2.2.3.1
$\mathfrak{T}_{add}$	Edge Addition Operation	Sec.2.3.1
$\Pi_{\ell \leq 2}(u, v)$	Set of Simple Directed Paths from u to v with length ≤ 2	Sec.2.3.1
L	Length of a cycle or path	Sec.2.3.1
γ	Geometric Quantum (Directed 3-Cycle)	Sec.2.3.3
$\Phi(G)$	Lexicographic Potential $(L_{\max}, N_{L_{\max}})$	Sec.2.3.5
$L_{\max}$	Length of the longest simple cycle in G	Sec.2.3.5
$N_{L_{\max}}$	Count of distinct cycles of length $L_{\max}$	Sec.2.3.5
$\mathcal{R}$	The Rewrite Rule (Edge addition mechanism)	Sec.2.4.2
C	A Simple Directed Cycle	Sec.2.4.3
$\text{dist}_C(u, v)$	Distance between vertices along a cycle C	Sec.2.4.3.1
$\mathcal{O}_{add}$	Composite Addition Phase (Chord insertion)	Sec.2.4.5
$\mathcal{O}_{del}$	Composite Deletion Phase (Entropic breakage)	Sec.2.4.5
$\mathcal{S}_{step}$	Composite Update Step $(\mathcal{O}_{del} \circ \mathcal{O}_{add})$	Sec.2.4.5
$\leq$	Effective Influence Relation (Strict Partial Order)	Sec.2.6.2
$H(e)$	History Timestamp (Local relational time / discrete proper time)	Sec.2.6.2
π_{uv}	A specific Simple Directed Path instance from u to v	Sec.2.6.2
$\neg$	Logical negation	Sec.2.7.1
N	Total number of vertices in the graph	Sec.2.7.2
R	Radius of local computational patch	Sec.2.7.3
t_{crit}	Critical time where cycle diameter exceeds horizon	Sec.2.7.3
ρ	Edge density of the graph	Sec.2.7.3.1
D	Graph Diameter	Sec.2.7.5
E_{sync}	Energy required for global synchronization	Sec.2.8
ΔF	Change in Free Energy	Sec.3.1

Symbol	Description	Context / First Used
r	The Root Vertex ($d_{in}(r) = 0$)	Sec.3.1.2
$d(v)$	Logical Depth of vertex v from root	Sec.3.1.2
$\pi(v)$	Parity of vertex v ($d(v) \pmod 2$)	Sec.3.1.2
V_{even}, V_{odd}	Vertex partitions based on depth parity	Sec.3.1.2
G_0	The Initial State (Vacuum) at $t_L = 0$	Sec.3.1.3
V_0, E_0	Vertex and Edge sets of the Initial State	Sec.3.1.3
$\mathrm{Aut}(G)$	Automorphism group of graph G	Sec.3.1.8
G_0	Regular Bethe Fragment (pre-geometric graph vacuum substrate)	Sec.3.2.1
k_{deg}	Internal coordination number (Regular Bethe Fragment)	Sec.3.2.1.1
$\mathcal{O}(G;\lambda)$	Structural Optimality Score	Sec.3.2.9
$H_S(G)$	Shannon entropy of the orbit size distribution	Sec.3.2.9
λ	Weighting parameter for optimality score	Sec.3.2.10
$\mathbf{A}$	Annotation structure (a_V, a_E)	Sec.3.3.1
φ	An automorphism mapping	Sec.3.3.1
$\mathfrak{G}_{\text{sym}}$	Formal Symmetry Framework (relational automorphism group)	Sec.3.3.2
$\mathcal{S}_{\text{sites}}(G)$	Set of candidate rewrite sites	Sec.3.3.3
e_{tunnel}	Symmetry-breaking tunneling edge	Sec.3.4.2
$\mathcal{T}_{\text{tunnel}}$	Tunneling Operator	Sec.3.4.2.1
d_H	Hamming Distance	Sec.3.4.2.1
$\chi(G)$	Chromatic Number	Sec.3.4.2.1
ϵ_{geo}	Geometric Self-Energy	Sec.3.4.5
$\mathbb{P}_{\text{ign}}$	Probability of ignition (tunneling)	Sec.3.4.5
$\mathcal{H}$	Configuration Hilbert Space $(\mathbb{C}^2)^{\otimes K}$	Sec.3.5.1
$\mathcal{C}$	QECC Codespace (Protected subspace)	Sec.3.5.1
$\bar{d}(u, v)$	Undirected shortest-path metric	Sec.3.5.1
Π_{cycle}	Hard Constraint Projector (2-Cycle)	Sec.3.5.1
Π_{local}	Projector enforcing locality distance	Sec.3.5.1
Z_{uv}	Pauli-Z operator on edge qubit (Check)	Sec.3.5.1
K_{uv}	Geometric Check Operator (Triplet stabilizer)	Sec.3.5.1
λ_{uv}	Syndrome eigenvalue (± 1)	Sec.3.5.1
X_{uv}	Pauli-X operator on edge qubit (Action)	Sec.3.5.3.2
$\mathbf{Caus}_t$	Internal Causal Category (Path Category)	Sec.4.1.1

Symbol	Description	Context / First Used
Hist	Global Historical Category (Embeddings)	Sec.4.1.2
AnnCG	Category of Annotated Causal Graphs	Sec.4.3.1
R_T	Awareness Endofunctor	Sec.4.3.2
σ_G	Freshly computed syndrome map	Sec.4.3.2
ϵ	Counit (Context Extraction)	Sec.4.3.3
δ	Comultiplication (Meta-Check)	Sec.4.3.4
T_c, β_c	Vacuum Temperature / Base Conversion Modulus ($\ln 2$)	Sec.4.4.1
ΔS_{close}	Entropy of Closure ($\ln 2$)	Sec.4.4.3
ε_{geo}	Microscopic Channel Self-Energy ($\frac{\ln 2}{3} \approx 0.231049$)	Sec.4.4.4
Λ_{theory}	Simplicial Permittivity / Cosmological Vacuum Drive ($2^{-6} = 0.015625$)	Sec.4.4.5
λ_0	Catalytic Defect Relaxation Generator ($e - 1 \approx 1.7183$)	Sec.4.4.6
μ_0	Modular S-Duality Friction Coefficient ($1/\sqrt{2\pi} \approx 0.3989$)	Sec.4.4.7
τ_{homeo}	Operational Homeostatic Stopping Time	Sec.4.5.1
$\mathcal{R}$	Universal Constructor (Rewrite Rule)	Sec.4.5.1
$\chi(\sigma_e)$	Catalytic Tension Factor	Sec.4.5.2
P_{acc}	Addition Proposal Probability	Sec.4.5.3
P_{del}	Deletion Proposal Probability	Sec.4.5.4
$\mathbb{P}_{\text{acc}}$	Acceptance Probability (Addition)	Sec.4.5.6
$\mathbb{P}_{\text{del}}$	Acceptance Probability (Deletion)	Sec.4.5.7
$\mathcal{U}$	Universal Evolution Operator ($\mathcal{D} \circ \mathcal{M} \circ \mathcal{P}_{\text{prop}} \circ \mathcal{A}$)	Sec.4.6.1
Σ_{valid}	State space of axiomatically compliant graphs	Sec.4.6.1
$\mathcal{A}$	Awareness Diagnostic Mapping	Sec.4.6.1
$\mathcal{P}_{\text{prop}}$	Stochastic Proposal Kernel	Sec.4.6.1
$\mathcal{M}$	Addition Merge Map	Sec.4.6.1
$\mathcal{D}$	Deletion Excision Map	Sec.4.6.1
$\mathbb{P}(G \to G')$	Transition Probability	Sec.4.6.3
Ω_N	Cardinality of configuration space on N vertices	Sec.5.1.1
$S(N)$	Total Entropy ($c \cdot N$)	Sec.5.1.1
c_{cap}	Specific entropy per event (Capacity)	Sec.5.1.1
$I(R_A; R_B)$	Mutual Information between disjoint regions	Sec.5.1.2
ξ	Correlation Length (Entropic decay scale)	Sec.5.1.2
V_ξ	Correlation Volume ($V \propto \xi^3$)	Sec.5.1.2.2
J_{in}, J_{out}	Topological Fluxes (Creation/Deletion)	Sec.5.2

Symbol	Description	Context / First Used
$N_3(t)$	Population of 3-cycles (Geometric Quanta)	Sec.5.2.1
$\rho(t)$	Normalized 3-cycle density (N_3/N)	Sec.5.2.2
Λ_0	Vacuum Permittivity (Ignition Flux)	Sec.5.2.3
d_c	Upper critical dimension ($d = 4$)	Sec.5.2.4.2
μ	Geometric Friction Coefficient ($1/\sqrt{2\pi}$)	Sec.5.2.5
λ_{cat}	Catalysis Coefficient ($e - 1$)	Sec.5.2.6
$\mathcal{V}_{\text{phys}}$	Region of Physical Viability (equilibrium parameter manifold)	Sec.5.3.1
$\mathcal{P}_{\text{sweep}}$	Parameter Sweep Protocol (systematic vacuum calibration)	Sec.5.3.2
$\mathcal{C}_{\text{viab}}$	Viability Channel (stable topological evolution trajectory)	Sec.5.3.4
ρ^*	Equilibrium 3-cycle density (≈ 0.03)	Sec.5.4.1
$F(\rho)$	Net Flux Function ($J_{in} - J_{out}$)	Sec.5.4.2.1
J	Jacobian Eigenvalue (Stability indicator)	Sec.5.4.4.1
$\bar{d}(u, v)$	Undirected shortest-path metric	Sec.5.5.2
$\langle k \rangle$	Mean vertex degree	Sec.5.5.3
$D_{\max}$	Maximum vertex degree bound	Sec.5.5.3
$K(u, v)$	Causal Ollivier-Ricci curvature	Sec.5.5.4
$W_1(\mu_u, \mu_v)$	Wasserstein-1 Distance	Sec.5.5.4.1
C_{cov}, γ	Covariance amplitude and decay rate	Sec.5.5.5
C_k	Count of simple cycles of length k	Sec.5.5.6
$B(v, r)$	Volume of geodesic ball of radius r	Sec.5.5.7
R	Causal Horizon (Radius of local operator)	Sec.6.1.1
$V_\xi(t)$	Jones Polynomial of subgraph ξ	Sec.6.1.1
G_t^*	Geometric vacuum at homeostatic fixed point	Sec.6.1.2
ρ^*	Equilibrium 3-cycle density (≈ 0.03)	Sec.6.1.2
$\rho(\zeta)$	Local 3-cycle density of excitation	Sec.6.1.2
$\mathcal{C}$	QECC Codespace (Protected subspace)	Sec.6.1.2
$w(\zeta)$	Writhe of the configuration	Sec.6.1.2
L_{ij}	Pairwise Linking Number	Sec.6.1.2
σ	Syndrome value (± 1)	Sec.6.1.2
J_{in}, J_{out}	Topological Fluxes (Creation/Deletion)	Sec.6.1.2
$\chi(\sigma)$	Catalytic Tension Factor	Sec.6.1.2
$\mathfrak{T}$	Elementary Task Space	Sec.6.1.3.1
$\mathbb{P}_{del}$	Deletion Probability	Sec.6.1.4
Inv	Generic topological invariant	Sec.6.1.5
β_n	Braid on n ribbons	Sec.6.2.1
$\mathfrak{su}(n)$	Special Unitary Lie Algebra	Sec.6.2.1

Symbol	Description	Context / First Used
$A(n)$	Anomaly Coefficient	Sec.6.2.1
$C[\beta]$	Minimal Crossing Number	Sec.6.2.1
f^{abc}	Structure constants of Lie algebra	Sec.6.2.2.1
B_n	Braid Group on n strands	Sec.6.2.5.1
C_C	Crossing Complexity	Sec.6.3.1
C_T	Torsional Complexity	Sec.6.3.2
m	Topological Mass (Informational Inertia)	Sec.6.3.3
k_{writhe}	Mass-Writhe coupling constant	Sec.6.3.3
N_3	Count of 3-cycles (Geometric Quanta)	Sec.6.3.4
k_c	Crossing proportionality constant	Sec.6.3.4
k_t	Torsional proportionality constant	Sec.6.3.7
$\mathcal{B}_{\text{lin}}$	Linear Barrier (density growth containment threshold)	Sec.6.4.1
Σ	Sequence of rewrite operations	Sec.6.4.4.1
Ξ	Set of all localized excitations	Sec.6.4.5
ζ	Localized excitation (subgraph of G_t^*)	Sec.6.5
b_i	Braid group generator	Sec.6.5
L_S	Spin Operator (Product of rung Z-operators)	Sec.7.1.1
Z_{e_i}	Pauli-Z operator on rung edge e_i	Sec.7.1.1
$\hat{P}_{12}$	Particle Exchange Operator	Sec.7.1.2
s	Spin quantum number (1/2)	Sec.7.1.2
ϕ	Topological phase factor (-1)	Sec.7.1.2
Π_s	Spin Projector	Sec.7.1.2
$\hat{\mathcal{T}}$	Unitary Twist Operator	Sec.7.1.3
$\|\psi_{violation}\rangle$	State of dual fermion occupancy (Forbidden)	Sec.7.2.4
Π_{cycle}	Hard Constraint Projector (2-Cycle)	Sec.7.2.4
Q	Electric Charge Operator	Sec.7.3.1
$w(\beta)$	Total Writhe of braid β	Sec.7.3.1
Q_ν, Q_e	Charge of neutrino (0), electron (-1)	Sec.7.3.5.1
$C(w)$	Topological Complexity (Sum of absolute writhes)	Sec.7.3.5.1
Q_d, Q_u	Charge of down quark ($-1/3$), up quark ($+2/3$)	Sec.7.3.6.1
k	Charge normalization constant (1/3)	Sec.7.3.7
Y	Hypercharge	Sec.7.3.7.2
m	Topological Mass (Informational Inertia)	Sec.7.4.1
N_3	Count of 3-cycles (Geometric Quanta)	Sec.7.4.1
κ_m	Universal Mass Constant (≈ 0.170 MeV)	Sec.7.4.2
U_{braid}	Internal Energy (Topological)	Sec.7.4.3
S_{braid}	Configurational Entropy (Zero)	Sec.7.4.3

Symbol	Description	Context / First Used
ϵ_{geo}	Geometric Self-Energy	Sec.7.4.3.1
k_{share}	Geometric Sharing Integer (1)	Sec.7.4.5
$\mathcal{R}$	Unitary Rewrite Operator ($e^{i\hat{H}}$)	Sec.8.1.1
$\hat{H}_i$	Hamiltonian Generator for rewrite $\mathcal{R}_i$	Sec.8.1.1
f_{ijk}	Structure Constants of the Lie algebra	Sec.8.1.1
σ_i	Braid Group Generator (swap ribbons $i, i+1$)	Sec.8.1.2
J^μ	Weak Current	Sec.8.1.2.1
$\lambda^{(i,j)}$	Traceless Hermitian Basis Matrix	Sec.8.2.1
G_{ab}	Gram Matrix element	Sec.8.2.5.1
γ^5	Chirality Operator	Sec.8.2.7
$\mathcal{R}_W$	Flavor-Changing Rewrite Process (Weak)	Sec.8.3.1
χ	Chiral Invariant (Sign of timestamp difference)	Sec.8.3.1
θ_W	Electroweak mixing angle (Weinberg angle)	Sec.8.4.1
p_3, p_4	Probabilities of 3-cycle and 4-cycle rewrites	Sec.8.4.1
$\sin^2\theta_W^{(0)}$	Isolated doublet rewrite baseline ($= 0.25$)	Sec.8.4.5
g	SU(2) Gauge Coupling Constant	Sec.8.5.1
α_{topo}	Topological Energy Scale ($\ln 2/4$)	Sec.8.5.1
M	Vertex Multiplicity Factor ($M = 7$)	Sec.8.5.6
v	Higgs Vacuum Expectation Value (VEV)	Sec.8.6.1
y_f	Yukawa Coupling for fermion f	Sec.8.6.5
N_{scale}	Vacuum Characteristic Quantum Supply	Sec.8.6.5
m_W, m_Z	Masses of W and Z Bosons	Sec.8.7
G_{GUT}	Candidate Grand Unified Theory group	Sec.9.1.2
$r(G)$	Rank of a Lie algebra	Sec.9.1.2.1
$\sin^2\theta_W(M_{\text{GUT}})$	Grand Unified $SU(5)$ boundary value ($= 3/8 = 0.375$)	Sec.9.1.5
$A(R)$	Anomaly Coefficient	Sec.9.1.5
$\bar{\mathbf{5}}, \mathbf{10}$	SU(5) Representations	Sec.9.1.5
$\mathcal{R}_5$	Penta-Ribbon (5-strand topological composite state)	Sec.9.2.1
$V(C)$	Topological complexity potential landscape	Sec.9.3.1
ΔC	Complexity gap (Barrier height)	Sec.9.3.4.1
$\mathcal{R}_{LQ}$	Rewrite process for leptoquarks	Sec.9.4.1
$\hat{\lambda}_{LQ}$	Leptoquark generator	Sec.9.4.2
β_5	Penta-ribbon braid (Unified State)	Sec.9.4.4.1
C_{total}	Total topological complexity	Sec.9.4.4.1
L_{CW}	Linking number between Color and Weak sectors	Sec.9.4.4.1

Symbol	Description	Context / First Used
τ_p	Proton lifetime	Sec.9.5.2
S_{inst}	Instanton Action (Tunneling)	Sec.9.5.4
β_+, β_-	Braid and anti-braid segments (Folded)	Sec.9.6.1
m_D	Dirac mass term	Sec.9.6.2
M_R	Heavy right-handed neutrino mass	Sec.9.6.2
m_ν	Light neutrino mass	Sec.9.6.2
$N_{3,\max}$	Maximum sustainable complexity (Criticality)	Sec.9.6.7
M_{Pl}	Planck mass	Sec.9.6.8
$0_L\rangle, 1_L\rangle$	Logical qubit basis states (Ground/Excited)	Sec.10.1.1
β_e, β_{e*}	Physical electron braid topologies (Symmetric/Asymmetric)	Sec.10.1.1
$\hat{T}_{ij}$	Writhe Exchange Operator (Twist transfer)	Sec.10.1.5.1
$\hat{H}_X$	Hamiltonian for the Logical X transition	Sec.10.1.5.1
$\hat{C}^2_{SU(3)}$	Quadratic Casimir Operator (Color measurement)	Sec.10.1.6.1
$S^{(uvw)}_{\text{geom}}$	Geometric check operator (Z-type on cycles)	Sec.10.2.1
$S^{(k,i)}_{\text{ribbon}}$	Ribbon integrity operator (Z-type on ladders)	Sec.10.2.1
S^X_v	Vertex stabilizer (X-type flow check)	Sec.10.2.1
$\mathcal{S}$	The Stabilizer Group	Sec.10.2.2
$\mathcal{C}_L$	Logical Codespace (Protected subspace)	Sec.10.3.1
d	Code Distance ($d = 3$)	Sec.10.3.4
$\mathcal{R}_X$	Logical X gate rewrite process (Writhe Shuffle)	Sec.10.4.1
$\mathcal{R}_Z$	Logical Z gate rewrite process (QND Measure)	Sec.10.5.1
$\mathcal{R}_H$	Hadamard gate rewrite process (Thermo-Quench)	Sec.10.6.1
T_{local}	Local temperature of a graph region	Sec.10.6.2.1
$\mathcal{R}_{CZ}$	Controlled-Z gate rewrite process (Catalytic)	Sec.10.7.1
σ_{eff}	Effective stress syndrome	Sec.10.7.2.1
$\mathcal{R}_T$	T-gate rewrite process (Self-Braiding)	Sec.10.8.1
$\mathcal{C}_{QBD}$	Ribbon Category of stable braids	Sec.10.8.3
$\mathcal{G}_{phys}$	Universal Physical Gate Set	Sec.10.8.8
$\hat{D}$	Dehn Twist Operator	Sec.10.8.9
d_{GHW}	Gromov-Hausdorff-Wasserstein metric	Sec.11.1.1
π	Transport coupling (joint measure)	Sec.11.1.1

Symbol	Description	Context / First Used
d_{GHW}	Gromov-Hausdorff-Wasserstein metric on graph state space	Sec.11.1.1
$d_{GH}(X, Y)$	Gromov-Hausdorff distance	Sec.11.1.1.1
$d_H(A, B)$	Hausdorff distance	Sec.11.1.1.1
$W_1(\mu_X, \mu_Y)$	Wasserstein-1 transport metric	Sec.11.1.1.1
$\bar{d}(u, v)$	Undirected shortest-path metric	Sec.11.1.2
$N^+(u), N^-(u)$	Future/Past causal neighborhoods	Sec.11.2
α	Laziness parameter (self-mass)	Sec.11.2
β	Neighborhood mass parameter $((1-\alpha)/2)$	Sec.11.2.1
μ_u	Lazy causal probability measure for Ollivier-Ricci curvature	Sec.11.2.1
μ_u	Lazy causal probability measure for vertex u	Sec.11.2.1.1
$\mathbb{I}[\cdot]$	Indicator function	Sec.11.2.1.1
$K(u, v)$	Causal Ollivier-Ricci curvature	Sec.11.2.2
$H(\mu_u)$	Shannon entropy of measure μ_u	Sec.11.2.3
$h(\alpha)$	Allocation entropy function	Sec.11.2.3.1
d_{dir}	Directed distance function (shown insufficient)	Sec.11.2.4.1
$\mathcal{S}_{\mathrm{EH}}$	Discrete Einstein-Hilbert Action on causal graphs	Sec.11.3.1
$\Delta\mathcal{S}$	Variation in total action	Sec.11.3.2
K_{baseline}	Baseline curvature in sparse graph	Sec.11.3.2.1
m_w	Zero-cost shared mass at vertex w	Sec.11.3.3.1
$\tilde{\mathcal{L}}_t$	Consistently weighted graph Laplacian	Sec.12.1.1
$-\Delta_g$	Laplace-Beltrami operator	Sec.12.1.2
$\tilde{\lambda}_k^{(t)}$	Eigenvalues of $\tilde{\mathcal{L}}_t$	Sec.12.1.3
$\psi_k^{(t)}$	Eigenfunctions of $\tilde{\mathcal{L}}_t$	Sec.12.1.3
f_k	Continuum eigenfunctions	Sec.12.1.3.1
$p_t(x, y)$	Heat kernel on graph/manifold	Sec.12.1.4
$\widetilde{\mathcal{G}}_{ij}^{(t)}$	Coarse-grained (averaged) Einstein tensor	Sec.12.2.1
$\widetilde{T}_{ij}^{(t)}$	Coarse-grained (averaged) stress-energy tensor	Sec.12.2.1
$\hat{n}_e$	Unit direction vector of edge e	Sec.12.2.1
$B(x, R)$	Mesoscopic ball of radius R	Sec.12.2.1
κ'	Continuum gravitational coupling constant	Sec.12.2.5
$\Delta_{\tilde{\mathcal{L}}}$	Spectral manifold reconstruction operator	Sec.12.3.1
T_{ab}	Discrete stress-energy tensor	Sec.13.1.1
$P_{\mathrm{add}}(a, b)$	Probability of edge addition	Sec.13.1.1
$P_{\mathrm{del}}(a, b)$	Probability of edge deletion	Sec.13.1.1
$\mathbb{E}[\Delta \deg(a)]$	Expected degree change	Sec.13.1.2.1
ℓ_0	Microscopic discreteness / Planck area element	Sec.13.1.5.1
κ	Discrete gravitational coupling	Sec.13.2

Symbol	Description	Context / First Used
$\mathcal{G}_{ab}$	Discrete Einstein Tensor quantifying local curvature response	Sec.13.2.1
$\mathcal{G}_{ab}$	Discrete Einstein tensor	Sec.13.2.1.1
R_{disc}	Discrete scalar curvature	Sec.13.2.1.1
$\mathcal{S}[G]$	Discrete Einstein-Hilbert action	Sec.13.2.3
$\mathcal{T}_{\text{conv}}$	Topological convergence operator on causal graphs	Sec.13.3.1
M	Continuous Lorentzian manifold	Sec.14.1.1
$g_{\mu\nu}$	Lorentzian spacetime metric tensor	Sec.14.1.1
N	Lapse function (coordinate update rate)	Sec.14.1.2
N^i	Shift vector (coordinate spatial offset)	Sec.14.1.2
$\|\Psi\rangle$	Wavefunction of the universe	Sec.14.1.4.1
$\mathcal{H}_{\text{rel}}$	Relational Hamiltonian constraint operator	Sec.14.2.1
$\hat{H}$	Hamiltonian constraint operator	Sec.14.3.1
Λ	Cosmological constant	Sec.14.3.1
$T_{\mu\nu}$	Continuous stress-energy tensor	Sec.14.4.2
G	Emergent gravitational constant	Sec.14.4.3
K_{ij}	Extrinsic curvature tensor	Sec.14.6
S_{EE}	Entanglement entropy	Sec.14.6
$S(A)$	boundary entanglement entropy of region A	Sec.15.1.1
ρ_A	Reduced density matrix of region A	Sec.15.1.1
E_{bridge}	Entanglement shortcut edges (non-local)	Sec.15.1.1
d_{geo}	Emergent spatial distance on manifold	Sec.15.1.2
d_{topo}	Intrinsic topological distance on causal graph	Sec.15.1.2
$W_1(\mu_X, \mu_Y)$	Wasserstein-1 transport metric	Sec.15.1.2
$\mathcal{S}$	Stabilizer group protecting codespace	Sec.15.1.4
$\|\Psi\rangle$	Wavefunction of the universe	Sec.15.1.4.1
E_{bulk}	Standard spatial edges (local)	Sec.15.1.5.1
S	Bell CHSH correlation metric	Sec.15.2.1
$\mathcal{E}_\Gamma$	Causal history path ensemble	Sec.15.4.1
$\mathcal{TN}$	Causal Tensor Network (Renormalization flow)	Sec.16.1.1
ℓ_0	Microscopic discreteness / Planck area element	Sec.16.1.1
$S(A)$	boundary entanglement entropy of region A	Sec.16.1.2
γ_A	Ryu-Takayanagi minimal bulk surface	Sec.16.1.2
G_N	Boundary Newton gravitational constant	Sec.16.1.2

Symbol	Description	Context / First Used
W_k	Isometric tensor mapping bulk to boundary	Sec.16.1.5.1
ρ_{max}	Maximum bulk informational capacity density	Sec.16.2.1
$I(R)$	Information bound of spatial region R	Sec.16.2.2
$S(\Omega)$	Universal holographic information bound on spatial subgraph	Sec.16.2.2
A	Area of black hole horizon / holographic screen	Sec.16.2.4
Φ_S	Information entropy flux through holographic screen boundary	Sec.16.2.4
$\mathcal{TN}_{\text{bulk}}$	Bulk causal tensor network geometry	Sec.16.3.1
$\mathcal{A}_{\text{RT}}$	Ryu-Takayanagi holographic entanglement area	Sec.16.4.1
$B_{\mu\nu}$	Kalb-Ramond 2-form field	Sec.17.1
A_μ	Emergent heterotic gauge field	Sec.17.1
Σ	Discrete worldsheet / causal tube	Sec.17.1.1
Φ	Dilaton field	Sec.17.1.4.1
S_{NG}	Nambu-Goto informational action	Sec.17.1.5
T_0	Relativistic string tension	Sec.17.1.5
R	Wess-Zumino compactification radius	Sec.17.2.1
$H(R)$	Hamiltonian operator of compactified string	Sec.17.2.2
T	T-duality mapping operator	Sec.17.2.2
D_L, D_R	Left-moving and right-moving critical dimensions	Sec.17.3
$\mathcal{S}_{\text{WS}}$	Causal worldsheet action in heterotic string target space	Sec.17.4.1
$E_8 \times E_8$	Heterotic unified gauge lattice group	Sec.17.4.2
$g_{\mu\nu}$	Lorentzian spacetime metric tensor	Sec.17.4.2
G_0	Pre-Geometric Vacuum (3-regular directed bipartite Bethe tree substrate)	Sec.18.1.1
V_A, V_B	Bipartite vertex partition of the pre-geometric tree vacuum	Sec.18.1.1
ρ_3	Directed 3-cycle intensive density in pre-geometric vacuum ($= 0$)	Sec.18.1.1
d_S, d_H	Spectral and Hausdorff dimensions of pre-geometric vacuum ($= 1$)	Sec.18.1.1
$\mathcal{U}$	Universal Evolution Operator initiating dynamical graph sequencer	Sec.18.1.1
P_{nuc}	Spontaneous loop nucleation tunneling probability	Sec.18.1.2
Γ_{spark}	Primordial loop nucleation ignition current / spark rate	Sec.18.1.2

Symbol	Description	Context / First Used
Λ	Spontaneous 3-cycle creation seed rate ($= 0.0156$)	Sec.18.1.2
k, b	Root coordination number ($k = 3$) and internal branching factor ($b = 2$)	Sec.18.1.3
$\mathcal{P}_{\text{prec}}$	Precursor 2-path abundance for 3-cycle closure	Sec.18.1.4
$\hat{P}$	Topological parity projection operator breaking bipartite stasis	Sec.18.1.5
λ_J	Linearized Jacobian negative feedback eigenvalue (< 0)	Sec.18.1.7
$N_3(t)$	Directed 3-cycle count (emergent macroscopic spatial 3-volume)	Sec.18.2.1
$a(t)$	Emergent cosmic scale factor ($a(t) = CN_3(t)^{1/3}$)	Sec.18.2.1
$H(t)$	Discrete Hubble expansion parameter ($H(t) \equiv \frac{1}{3}\frac{\dot{N}_3}{N_3}$)	Sec.18.2.2
$\rho(t)$	Intensive 3-cycle density ($N_3(t)/\|V(t)\|$)	Sec.18.3.1
$F(\rho)$	Master Equation intensive rate function ($(\Lambda + 9\rho^2)e^{-6\mu\rho} - 0.5\rho$)	Sec.18.3.1
λ_{cat}	Catalytic deletion coupling parameter	Sec.18.3.1
μ	Steric hindrance / homeostatic friction parameter ($= 0.399$)	Sec.18.3.2
ρ_0	Frictionless cycle growth threshold density ($> 1/18 \approx 0.0556$)	Sec.18.3.2
ρ_c	Exact unpumped critical nucleation barrier ($= \frac{1}{24-6e} \approx 0.130$)	Sec.18.3.2.1
ρ^*	Homeostatic stable fixed-point attractor density (≈ 0.0370)	Sec.18.3.4
ε	First inflationary slow-roll parameter ($-\dot{H}/H^2 \approx 0.0125$)	Sec.18.4.1
η	Second inflationary slow-roll parameter ($-\ddot{\rho}/(H\dot{\rho}) \approx 0.0050$)	Sec.18.4.1
n_s	Primordial scalar spectral index ($= 1 - 2\varepsilon - 2\eta \approx 0.965$)	Sec.18.4.1
$P_{\mathcal{R}}(k)$	Primordial curvature perturbation power spectrum amplitude at horizon exit	Sec.18.4.1
$D_{\text{noise}}(\rho)$	Steric-damped stochastic update noise diffusion coefficient	Sec.18.4.3
σ_0^2	Bare quantum rewrite fluctuation amplitude	Sec.18.4.3.1
Ω_K	Spatial curvature density parameter (≈ 0, flat universe)	Sec.18.5.1
K_{spatial}	Emergent spatial Ricci curvature scalar	Sec.18.5.1

Symbol	Description	Context / First Used
d_{tree}	Logarithmic pre-geometric tree diameter ($\mathcal{O}(\ln N)$ solving horizon problem)	Sec.18.5.2
T_{rh}	Primordial reheating temperature generated by topological defect annihilation	Sec.19.1.1
ρ_{rh}	Energy density of graph vacuum at reheating	Sec.19.1.1
τ_{relax}	Steric density relaxation timescale	Sec.19.1.2
Γ_{def}	Topological defect nucleation rate per unit 3-volume	Sec.19.1.3
$P(C)$	Boltzmann creation probability for braid defect of crossing complexity C	Sec.19.1.4
ΔB	Net baryon number asymmetry generated during baryogenesis epoch	Sec.19.2.1
χ_{CP}	Topological CP-violating phase angle in braid chiral vertex updates	Sec.19.2.2
$\mathcal{M}_{\text{hadron}}$	Hadronic mass matrix functional on braid defect spectrum	Sec.19.3.1
Δm_{np}	Neutron-proton rest mass difference (= 1.2930 MeV)	Sec.19.3.2
κ_{top}	Topological mass splitting energy scale (= 0.6843 MeV)	Sec.19.3.2
Δm_{top}	Topological mass defect contribution (= +2.0530 MeV)	Sec.19.3.2
Δm_{EM}	Hadronic electromagnetic Coulomb self-energy contribution (−0.7600 MeV)	Sec.19.3.2
Y_p	Primordial Helium-4 mass fraction (≈ 0.245)	Sec.19.4.1
τ_n	Free neutron lifetime (= 879.4 s)	Sec.19.4.1
η_B	Baryon-to-photon ratio ($\eta_B \approx 6.1 \times 10^{-10}$)	Sec.19.4.1
T_f	Weak interaction freeze-out temperature ($T_f \approx 0.8135$ MeV)	Sec.19.4.2
B	Deuteron binding energy ($B = 2.224$ MeV)	Sec.19.4.3
n_B/n_γ	Baryon asymmetry parameter at Big Bang Nucleosynthesis freeze-out	Sec.19.4.3
z_{rec}	Cosmological recombination redshift ($z_{\text{rec}} \approx 1100$)	Sec.20.1.1
T_{rec}	Recombination temperature (≈ 0.3 eV)	Sec.20.1.1
τ_{erg}	Plasma ergodic mixing timescale in primordial baryon-photon fluid	Sec.20.1.2
β_3	3-strand stable baryon braid defect state	Sec.20.1.2.1

Symbol	Description	Context / First Used
$X_e(z)$	Free electron ionization fraction during recombination epoch	Sec.20.1.3
α_B	Case-B hydrogen recombination rate coefficient	Sec.20.1.3
Ω_Λ	Dark energy cosmological constant density parameter (≈ 0.685)	Sec.20.1.3.1
σ_8	Linear matter fluctuation amplitude within $8h^{-1}$ Mpc spheres (≈ 0.811)	Sec.20.1.3.1
$\Delta T/T$	Sachs-Wolfe gravitational potential temperature perturbation	Sec.20.1.4
$g(z)$	Photon decoupling visibility function ($g(z) = -\dot{\tau}e^{-\tau}$)	Sec.20.1.5
$\tau(z)$	Thomson scattering optical depth from observer to redshift z	Sec.20.1.5
C_ℓ	CMB angular power spectrum multipole expansion coefficients	Sec.20.2.1
ℓ_{peak}	First acoustic peak multipole location ($\ell_{\text{peak}} \approx 220$)	Sec.20.2.1
Θ_A	Sound horizon acoustic angular scale ($\Theta_A \approx 0.596^\circ$)	Sec.20.2.1
$r_s(z_{\text{dec}})$	Comoving sound horizon scale at photon decoupling	Sec.20.2.2
ρ_{crit}	Critical cosmological energy density ($3H_0^2/8\pi G$)	Sec.20.3.1
Ω_m	Total matter cosmic density parameter (≈ 0.315)	Sec.20.3.1
$w(z)$	Dark energy equation of state parameter ($w(z) = p/\rho \approx -1$)	Sec.20.4.1
$P(k)$	Linear matter power spectrum of cosmological density fluctuations	Sec.20.5.1
f_{NL}	Primordial non-Gaussianity amplitude parameter ($f_{\text{NL}} \ll 1$)	Sec.20.6.1
$H_1(\Sigma, \mathbb{Z})$	First boundary homology group protecting β_4 relic stability	Sec.21.1
β_4	4-strand gauge-sterile dark matter braid defect	Sec.21.1.1
Y_∞	Asymptotic dark matter comoving yield (n_{DM}/s)	Sec.21.1.2.1
$\text{Tr}(T^a)$	Vanishing gauge generator trace for sterile dark braid defects	Sec.21.1.3
m_{B_4}	Ground-state rest mass of 4-strand dark relic ($= 5.026$ GeV)	Sec.21.1.4
κ_H	Hadronic / crystallization scale mass quantum (≈ 314.159 MeV)	Sec.21.1.4
$C[\beta_4]$	Crossing complexity of 4-strand defect ($= 16$)	Sec.21.1.4
n_{B_4}	Volumetric number density of 4-strand dark relics	Sec.21.1.5

Symbol	Description	Context / First Used
ξ_{KZ}	Kibble-Zurek correlation length at vacuum crystallization	Sec.21.1.5
σ_{ann}	Topological relic annihilation cross section	Sec.21.1.6
Ω_{DM}	Cold dark matter cosmic density parameter (≈ 0.265)	Sec.21.1.6
T_{fo}	Dark matter relic freeze-out temperature	Sec.21.1.6
Λ_{eff}	Effective cosmological constant from residual defect network tension	Sec.21.2.1
T_{defect}	Topological defect line tension on causal graph	Sec.21.2.1
σ_{SI}	Spin-independent dark matter-nucleon scattering cross section	Sec.21.3.1
$\rho_{\text{halo}}(r)$	Dark matter galactic halo radial density distribution profile	Sec.21.4.1
$\mathcal{C}_{\text{core}}$	Saturated graph core induced subgraph at extreme compression	Sec.22.1.1
ρ_{sat}	Incompressible saturated graph core density fixed point (ρ^*)	Sec.22.1.1
K_{sat}	Saturated core discrete causal curvature bound	Sec.22.1.2
Γ_{core}	Steric exponentially suppressed core rewrite rate ($\propto e^{-6\mu\rho_{\text{sat}}}$)	Sec.22.1.3
$N(r)$	Emergent gravitational lapse function in regular black hole core	Sec.22.1.7
r_{core}	Quantum core radius of non-singular regular black holes	Sec.22.1.8
$\Delta\varphi_{\text{prec}}$	Relativistic perihelion precession angle of planetary orbits	Sec.22.2.1
θ_{defl}	Gravitational light deflection angle by compact mass	Sec.22.2.2
$\Delta t_{\text{Shapiro}}$	Shapiro gravitational time delay of relativistic signals	Sec.22.2.3
$\gamma_{\text{PPN}}, \beta_{\text{PPN}}$	Parametrized Post-Newtonian parameters ($\gamma_{\text{PPN}} = \beta_{\text{PPN}} = 1$)	Sec.22.2.4
Γ_{evap}	Boundary swap Hawking evaporation rate of quantum black hole core	Sec.22.3.1
$\mathcal{M}_{\text{deg}}$	Degenerate tripartite braid media codespace lattice	Sec.22.4.1
f_{GW}	Gravitational wave frequency of coalescing compact braids	Sec.22.4.1
Φ_{Ginzburg}	Macroscopic Cooper braid condensate complex order parameter	Sec.22.5.1
ξ_{GL}	Ginzburg-Landau coherence length in macroscopic braid condensate	Sec.22.5.2

Symbol	Description	Context / First Used
λ_{London}	London magnetic penetration depth in superconducting braid media	Sec.22.5.2
$\mathbf{A}_{\text{eff}}$	Effective electromagnetic vector potential in superconducting graph	Sec.22.6.1
$\hat{H}_{\text{Ryd}}$	Driven-dissipative Rydberg blockade Hamiltonian	Sec.23.1.1
R_b	Rydberg blockade exclusion radius	Sec.23.1.1
R_{fac}	Rydberg facilitation resonance radius	Sec.23.1.1
ρ^*_{Ryd}	Synthetic quasi-stationary vacuum density (≈ 0.037)	Sec.23.1.2
β_{DP}	Directed percolation critical order parameter exponent	Sec.23.1.4
S_v, S_p	Trivalent stabilizer vertex divergence and plaquette flux checks	Sec.23.2.1
$\hat{U}_{\text{synd}}$	Transpiled trivalent stabilizer syndrome extraction operator	Sec.23.2.1
Δt_{circ}	Minimal syndrome extraction circuit depth ($4\tau_{\text{gate}}$)	Sec.23.2.1
p_{th}	Pre-geometric multi-qubit fault-tolerance threshold (≈ 0.104)	Sec.23.2.2
ϵ_L	Logical qubit error rate per syndrome extraction cycle	Sec.23.2.2
$\hat{\Phi}(L,t)$	Discrete metric optical phase shift operator	Sec.23.3.1
$S_h(f)$	Holographic strain cross-power spectral density	Sec.23.3.1
$\Diamond_x$	Spacetime causal diamond of cavity optical path	Sec.23.3.3
$\lvert\Psi_{\text{bif}}\rangle$	Geodesically bifurcated center-of-mass spatial superposition	Sec.23.4.1
$\mathcal{F}_{\text{graph}}$	Inner product overlap of disjointly deformed graph states	Sec.23.4.1
Γ_{dec}	Discrete lapse desynchronization decoherence rate	Sec.23.4.2
E_Δ	Relational gravitational self-energy difference	Sec.23.4.3
$\mathcal{H}_{\text{phys}}$	Gauge-invariant physical state space of ribbon networks	Sec.24.1.1
$\mathcal{P}_{\text{gauge}}$	Global compact Lie group averaging projector	Sec.24.1.3
$\lvert\Omega\rangle$	Non-degenerate gauge-invariant vacuum state	Sec.24.1.5
Δ_{YM}	Non-perturbative Yang-Mills topological mass gap	Sec.24.2.1

Symbol	Description	Context / First Used
$C(K)$	Knot minimal crossing number invariant	Sec.24.2.2
κ	Dimensionless ribbon elastic deformation modulus	Sec.24.2.3
Λ_{YM}	Dynamically transmuted non-perturbative hadronic scale	Sec.24.3.1
$\mathcal{D}_b$	Real-space causal poset block-decimation operator	Sec.24.3.2
β_0	One-loop non-Abelian Callan-Symanzik beta function coefficient	Sec.24.3.4
$\mathcal{W}(R,T)$	Rectangular non-Abelian Wilson loop operator	Sec.24.4.1
σ	Linear quark confinement string tension	Sec.24.4.1
$\hat{R}_{\mathrm{snap}}$	Dynamical ribbon bisection string-breaking rewrite operator	Sec.24.4.5
R_c	Meson pair crossover saturation distance (≈ 1.2 fm)	Sec.24.4.6
Θ	Causal poset wedge reflection time-involution operator	Sec.24.5.3
$\hat{T}$	Discrete causal poset transfer matrix operator	Sec.24.5.4
R_{gap}	Dimensionless gap-to-string tension scaling ratio ($3\sqrt{\kappa}$)	Sec.24.6.1
$\mathcal{G}$	Universal relational causal graph substrate	Sec.25.1.1
$\mathrm{Aut}(\mathcal{G})$	Automorphism group of the relational causal network	Sec.25.1.2
$\hat{P}_{\mathcal{S}}$	Comonadic stabilizer projection operator	Sec.25.1.3
$\mathcal{H}_{\mathrm{univ}}$	Universal state space of the causal network	Sec.25.1.3
ΔS_{rel}	Relational entropy generated by discrete lapse desynchronization	Sec.25.1.4
$b_1(\mathcal{G})$	First Betti number of causal graph cycle homology	Sec.25.1.5
$\hat{\mathcal{J}}$	Global cycle-inversion duality operator	Sec.25.1.5
R_{min}	Minimum spatial scale under cycle-basis inversion ($\approx \ell_0$)	Sec.25.1.5
Σ_{QES}	Cosmological quantum extremal surface	Sec.25.1.5
S_{vac}	Ground-state vacuum code entropy per erased link ($= \ln 2$)	Sec.25.1.5
$C_{\mu\nu\rho\sigma}$	Conformal Weyl curvature tensor	Sec.25.1.5
T^{μ}_{μ}	Trace of the energy-momentum tensor in conformal epoch	Sec.25.1.5
$\mathcal{R}$	Space of microscopic local graph rewrite rules	Sec.25.2.1
p_{eff}	Effective error rate per graph rewrite step	Sec.25.2.2

Symbol	Description	Context / First Used
$\mathcal{O}$	Embedded localized braided observer subsystem	Sec.25.3
$\hat{\mathcal{W}}_{\text{loop}}$	Self-observing participatory cosmic closure operator	Sec.25.3

Document Status

Draft Version 0.3

DOI: 10.5281/zenodo.17981900

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